

A comprehensive review on coupled processes and mechanisms of soil-vegetation-hydrology, and recent research advances

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Abstract Research on the coupling of soil, vegetation, and hydrological processes is not only a research hotspot in disciplines such as pedology, ecohydrology and Earth system science but also important for achieving sustainable development. However, scientists from different disciplines usually study the coupling mechanism of soil-vegetation-hydrological processes at very different space and time scales, and the mechanistic connections between different scales are quite few. This article reviewed research advances in coupled soil-vegetation-hydrological processes at different spatial scales—from leaf stomata to watershed and regional scales—and summarized the spatial upscaling methods and modeling approaches of coupled soil-vegetation-hydrological processes. We identify and summarize the following coupling processes: (1) carbon-water exchange in leaf stomata and root-soil interface; (2) changes in soil aggregates and profile hydraulic properties caused by plant roots and water movement; (3) precipitation and soil moisture redistribution by plant canopy and root; (4) interactions between vegetation patches and local hydrological process; (5) links between plant community succession and soil development; and (6) links between watershed/regional water budget and vegetation phenology and production. Meanwhile, the limitations and knowledge gaps in the observations, mechanisms, scaling methods, and modeling approaches of coupled soil-vegetation-hydrological processes were analyzed. To achieve a deep integration of various coupling processes across different spatiotemporal scales, future work should strengthen multiscale, multifactor and multiprocess soil-vegetation-hydrology coupling observations and mechanism studies, develop new scaling methods, identify different feedback pathways, and take time-variable plant behavior and soil hydraulic properties into account during modeling.

Keywords Ecohydrological process, Soil process, Plant-soil feedback, Modeling approaches, Scaling method

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1. Introduction

The complex interactions between the atmosphere, hydrosphere, biosphere and lithosphere are major research objectives of Earth System Science (Milly, 2009; Steffen et al., 2020). The study of coupled soil-vegetation-hydrological processes can help us better understand the interaction between different spheres of the Earth system. Soil acts as a nexus connecting different spheres (Li, 2011), and vegetation growth and hydrological processes regulate the matter and energy exchange among these spheres; therefore, studying these processes can provide a solid foundation for understanding the surface Earth system.

Humans rely on soil, vegetation, and water resources in natural ecosystems to live. However, with the growth of the global economy and population, these three resources are at risk (Liu et al., 2021; Zhou and Zhou, 2021). According to the United Nations, more than 40% of the global population is affected by water shortages, 52% of agricultural land is affected by soil degradation, and 3.3 million hectares of forests disappeared between 2010 and 2015 (<https://www.un.org/sustainabledevelopment/>). Meanwhile, soil, vegetation, and water are inherently coupled across a wide range of spatial and temporal scales (Fatichi et al., 2016); changes in one of them can affect the other two—for example, afforestation may induce watershed runoff decline and soil drying (Zhang et al., 2018). In this context, understanding the coupling and/or trade-off mechanism of soil, vegetation, and hydrological processes can provide vital guidance for formulating sustainable development policies.

Studies on the coupling mechanism of soil-vegetation-hydrological processes started in the 1980s (Ehrenfeld et al., 2005) and have gradually become a research hotspot in many disciplines. However, ecologists, soil scientists, hydrologists and climate modelers usually study the coupling mechanism of soil-vegetation-hydrological processes at very different space and time scales, and the mechanistic connections between different scales are quite a few, limiting the knowledge integration of coupled soil-vegetation-hydrological processes in different disciplines. For example, classic hydrological tools can simulate short-term (e.g., year) streamflow and vegetation water consumption well; however, when the focus shifts toward long-term processes or large-scale (e.g., regional) phenomena, where the water cycle and vegetation cover are mutually influenced, the classic ecohydrology method may cause misleading assessments because it cannot capture the feedbacks between vegetation and the water cycle (Fatichi et al., 2016; Li Y et al., 2018). For a much longer time scale (e.g., century), the soil (pedogenic) process is tightly connected to the ecohydrological process, whereas relevant models are currently lacking (Van der Meij et al., 2018). How to mechanistically connect fast ecological and hydrological processes with slow soil processes, and how to

predict large-scale phenomena through small-scale processes remain challenging.

Given that two-way coupling among soil, vegetation, and hydrological processes does exist at various spatiotemporal scales, it is difficult to accurately simulate complex ecohydrological processes or land-climate feedback if coupled soil-vegetation-hydrological processes are not well represented within the Earth system (Fatichi et al., 2016). Therefore, this article aims to systematically review the existing studies in coupled soil-vegetation-hydrological process, outline the typical coupling process and mechanism, and summarize upscaling methods and modeling approaches, as well as their limitations and future directions, to provide a theoretical base for the advancement of Earth system science and the formulation of environmental policies.

2. Multiscale processes of coupled soil-vegetation-hydrology

Coupling, as a classic physical concept, has been widely used in Earth Science for describing the energy or matter exchange process produced by the interaction between two or more independent systems (Song et al., 2020). The coupling of soil-vegetation-hydrological processes largely depends on the scale. Phenomena observed or laws summarized on one scale are not always effective on another scale (Li et al., 2002). Therefore, in this article, we summarize six representative coupled soil-vegetation-hydrological processes at different spatial scales (Figure 1). The dominant coupling mechanism and its influencing factors change from leaf stomata to regional scales (Anderson et al., 2003).

2.1 Carbon-water exchange in leaf stomata and root-soil interface (10^{-5} – 10^{-3} m)

2.1.1 Carbon-water exchange mechanism in leaf stomata Stomata control carbon and water exchange between the leaf interior and the external atmosphere (Lawson and Vialet-Chabrand, 2019). In general, plants lose approximately 100–500 molecules of water to fix one molecule of CO_2 (Nobel, 2009); therefore, stomatal behavior is the key to the coupling of photosynthesis and transpiration, i.e., the key for vegetation-hydrological process coupling at the plant tissue scale (10^{-5} m) (Fatichi et al., 2016). In traditional research, the opening and closing behavior of stomata is usually described and predicted by plant hydraulic mechanisms (such as guard cell turgor pressure) and chemical mechanisms (such as abscisic acid concentration). In recent years, however, the “optimal theory” has become increasingly popular in ecohydrological modeling. The optimal theory of stomatal behavior assumes that the stomatal aperture is automatically

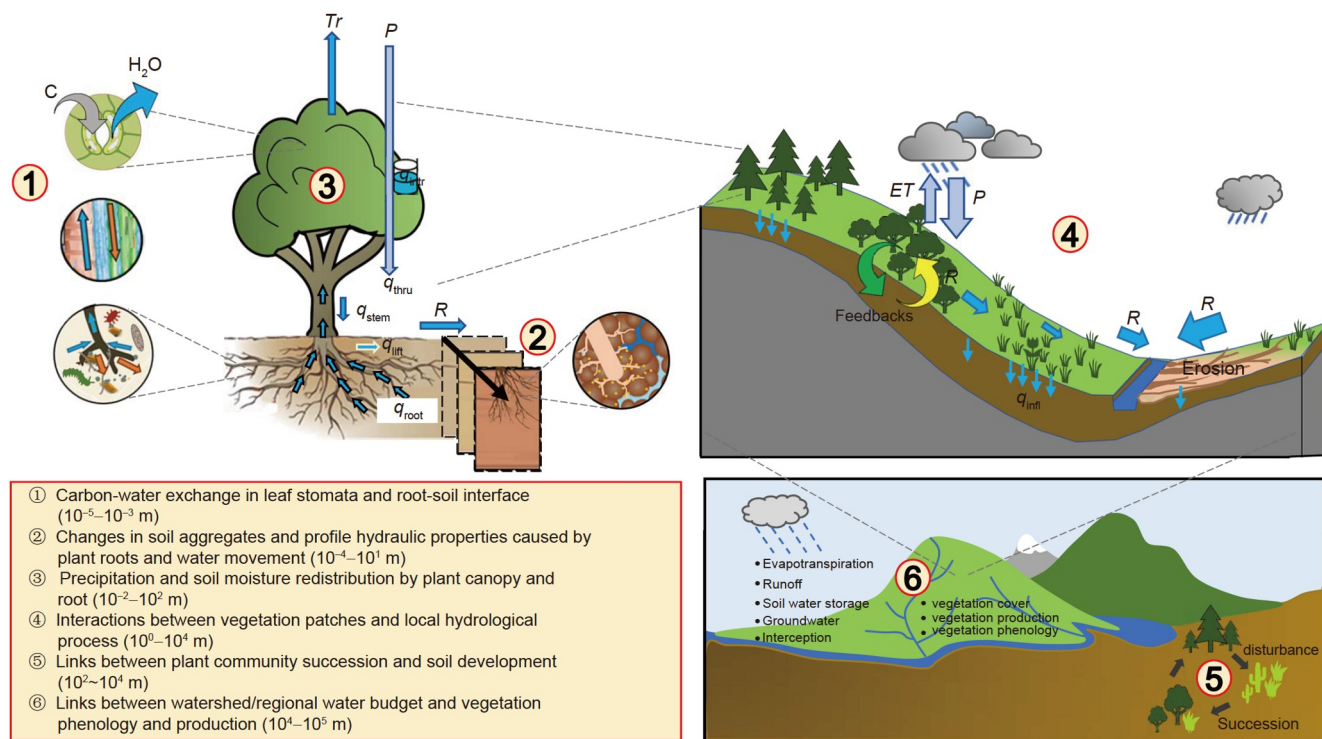


Figure 1 The coupling of soil-vegetation-hydrological processes at different spatial scales. The terms q_{intr} , q_{thru} , q_{stem} , q_{lift} , q_{root} , q_{infl} are plant canopy interception, throughfall, stemflow, hydraulic lift, water uptake by roots, and infiltration respectively. The terms Tr , P , ET , R are transpiration, precipitation, evapotranspiration and runoff, respectively.

adjusted according to the principle of “maximizing photosynthetic carbon fixation and minimizing transpiration water loss” (Cowan and Farquhar, 1977). Its concise assumption allows researchers to deduce analytical expressions of transpiration from optimal stomatal conductance functions (Katul et al., 2012) and thus employ many ecohydrological and Earth system models (ESM) (Buckley and Schymanski, 2013). Zhou et al. (2014) demonstrated that the optimal theory of stomatal behavior is, in fact, effective in various ecosystems when considering the VPD and time lags at the subdaily time scale. Future work should focus on linking the optimal theory and hydraulic regulation theory to better understand and model stomatal behaviors.

With the increase of spatial scale, the organized closure of many stomata across the leaf surface can significantly modify the temperature and humidity within the leaf boundary layer (Ball et al., 1987), and these modifications can feed back significantly on the behavior of the leaf stomata itself (Jarvis and McNaughton, 1986). When these feedback systems are not considered in upscaled models, leaf-level fluxes, and sensitivity to environmental changes can be misrepresented (Anderson et al., 2003). Besides, stomatal transpiration rate can also be limited by the xylem embolism (cavitation), caused by soil drought stress, freeze-thaw cycle, and plant diseases (Venturas et al., 2017). The embolization process is considered reversible, but the me-

chanism of embolization repair is still controversial (Jin et al., 2016). Overall, researches on stomata behaviors have made considerable progress; future work should pay more attention to stomatal behavior under the background of global change, such as elevated CO_2 (Fan et al., 2016).

2.1.2 Carbon, water, and nutrient flow at the root-soil interface

The soil environment around the root surface, i.e., the rhizosphere, is the primary habitat for many soil biota (Pierret et al., 2007; Faticchi et al., 2016). The “roots-rhizosphere-mycorrhizal-soil microorganism” system couples the “top-down carbon flow” with the “bottom-up water and nutrient flow”, making the rhizosphere interface the most active area of plant-soil interaction.

Specifically, plants transport their photosynthetic carbon uptake from leaves to underground; some carbon is used to support root growth—through increased root surface area, plants absorb a large amount of soil water and nutrients (Segal et al., 2008); the rest of carbon is released into the rhizosphere environment in the form of root exudates, which change the physical, chemical and biological soil properties of the rhizosphere so that plant roots can selectively recruit and support a large number of soil microorganisms there (Shen et al., 2021). The chemical matters released by the plant roots include H_2CO_3 in soil solution, organic acids, and

hydrogen ions, which strongly affect the soil weathering rates (Egli et al., 2008). Meanwhile, the compounds exuded from the plant roots, such as mucilage, enzymes, organic anions, and protons, greatly modify the soil chemical properties and increase the microbial community in the rhizosphere (Sasse et al., 2018). In turn, rhizosphere soil microorganisms improve plant nutrient utilization efficiency and maintain root health by promoting organic matter decomposition and nutrient mineralization, inhibiting harmful pathogenic fungi, and forming symbiotic mycorrhizae with roots (Liu W S et al., 2020; Wang S J, 2020). The decomposition of organic matter and litters by microbes can release nutrients such as N, P, S, etc., contributing to the “bottom-up nutrient flow” from soil to plants (Wang S J, 2020). It is estimated that 10% to 30% of the photosynthetic carbon produced by terrestrial plants is used to build root-fungal networks (i.e., mycorrhizal networks) in exchange for up to 90% of their nutrient requirements (Van der Heijden et al., 2015). Besides, some mycorrhizal networks can capture nutrients to plants from soil far away from roots, and allow rapid transfer of carbon and nutrients through advection and diffusion (Van der Heijden et al., 2015).

However, most previous studies have focused only on a few microbial species, such as *Rhizobium* and *Mycorrhizal*, and there is a lack of systematic research on the species and functions of other microorganisms involved in plant nutrient cycles as well as on the interaction mechanism between them and roots (Shen et al., 2021). Challenges in rhizosphere research involve understanding the integration of the multiple root-soil interaction processes occurring in the rhizosphere through noninvasive, real-time, and *in situ* investigations of local rhizosphere processes rather than simplified, uniform and often laboratory-based experimental tests (Wang X et al., 2020).

2.2 Changes in soil aggregates and profile hydraulic properties caused by plant roots and water movement (10^{-4} – 10^1 m)

2.2.1 Interactions between root activities, soil aggregates and soil pore water

Soil aggregates are the basis for constructing soil pore networks, and their formation process is strongly associated with root activities. The positive feedback process between soil aggregates and plant roots as well as root-associated microorganisms has long been recognized: first, plant root activities promote the formation of soil aggregates by enhancing the wet-dry cycle, exerting physical pressure, causing the presence of phenolic compounds in root exudates, and influencing the overall input of SOM; second, well-developed aggregates can improve soil water availability, enhance nutrient cycling, and have a potential positive effect on plant growth; finally, fast-growing plants further promote aggregate formation (Ehrenfeld et al., 2005;

Liu J Y et al., 2020). At the same time, some root-associated microorganisms, such as soil fungi, can promote aggregate formation by enhancing the cementation of soil particles and small-molecule organic matter, while the size of aggregates, in turn, affects the diversity of soil microbial communities (Huang et al., 2021).

In addition, the interaction between soil aggregate formation and soil pore water movement is nonnegligible when studying soil-hydrological process coupling at the soil pore scale. Soil pore water flow affects the formation, connectivity and stability of soil aggregates through water-associated clay translocation and wet-dry and freeze-thaw cycles (Van der Meij et al., 2018). On the other hand, the soil pore network, constructed by aggregates, controls the direction and velocity of soil pore water flow by regulating pore structure and properties (Alaoui et al., 2011). Small pores are conducive to water conservation, while large pores are conducive to water transport; the larger the pores between aggregates, the faster the pore water flow velocity. Changes in soil aggregates at the pore scale drive the dynamics of soil hydraulic properties at the profile scale; however, how to build mechanistic connections between them and how to predict profile-scale characteristics through pore-scale observations is still a huge challenge.

2.2.2 Changes in soil profile hydraulic properties caused by plant roots and water movement

Although profile soil hydraulic properties directly affect soil water movement in the short term and thereby affect plant growth (Wang C et al., 2020), in the long term, soil profile hydraulic properties are controlled by the root growing effect and water leaching at the profile scale, which is rarely recorded in field experiments or considered when modeling soil water dynamics over long-term periods (Alletto et al., 2015). Some previous work shows that the physical force of root growth on soil can lead to significant changes in soil saturated water content (θ_s) and saturated hydraulic conductivity (K_s) (Chen J Z et al., 2021): for example, Lu et al. (2020) suggested that fine roots (root diameter <1 mm) at low density easily block soil pores, leading to reduced θ_s and K_s ; fine roots at high density are prone to eliminate macropores while increasing pore volume, leading to increased θ_s and reduced K_s ; coarse roots (root diameter >2 mm) tend to cause small-scale compaction and macropore development, leading to increased θ_s and K_s . In addition, root growth can also change the soil bulk density and soil water retention curve of their respective profiles (Jotisankasa and Sirirattanakach, 2017). These effects cause the soil profile hydraulic characteristics to change dynamically within one growing season (Wang C et al., 2020), which is contrary to the traditional idea that soil properties are almost constant on an annual time scale. Therefore, improved knowledge of soil hydraulic property changes caused by plant roots is funda-

mental to accurately predict profile-scale soil processes.

The changes in soil profile hydraulic characteristics controlled by water movement are mainly chemical weathering, which may occur over millennial timescales, including decarbonization, clay translocation, podzolization, and gleization (Van der Meij et al., 2018). Taking clay translocation as an example, intuitively, the illuviation of clay in a soil horizon could decrease hydraulic conductivity by blocking soil pores. However, clay is also an important building material of soil aggregates, so the increase in clay content can also accelerate the formation of soil structure, with higher permeability as a result (Van der Meij et al., 2018). Boivin et al. (2004) believed that the increasing concentration of swelling clay minerals caused shrinkage and dry cracks of the illuvial horizon during the dry period, leading to a highly effective hydraulic conductivity. Van der Meij et al. (2018) suggested that the hydraulic conductivity first increased and then decreased through the cracking of the illuvial horizon. Equations describing soil chemical weathering that include water and soil moisture as drivers are often simplified and empirically based to reduce complexity and calculation demands (Egli and Fitze, 2001).

Soil profile hydraulic properties vary in space and time. However, in ecological and hydrological modeling studies, only a few researchers have considered the effect of root growth on soil hydraulic properties in short periods. Some researchers have built empirical relationships between plant biomass and soil permeability but lack a universal physical mechanism. In pedology modeling studies, it is still a challenge to construct an accurate soil forming function driven by annual water flow (Van der Meij et al., 2018); therefore, quantitatively describing the changes in soil hydraulic parameters over time is nearly impossible.

2.3 Precipitation and soil moisture redistribution by plant canopy and root (10^{-2} – 10^2 m)

2.3.1 Precipitation redistribution process by plant canopy

The plant canopy partitions rainfall into throughfall, stemflow and interception (Figure 1), which cause the redistribution of incident rainfall under the canopy and may affect the distribution pattern of soil moisture around the plant (Wang et al., 2016). In this process, the proportion of interception, throughfall, and stemflow are jointly determined by the plant canopy structure (such as crown width, plant height, leaf area, bark roughness, etc.), rainfall characteristics, and meteorological conditions (such as wind speed and direction) (Li, 2011; Yuan et al., 2016). In general, when the rainfall amount is identical, the larger the plant leaf area index (LAI), the higher the canopy interception ratio (Running and Coughlan, 1988). Stemflow is often inversely proportional to bark roughness and branch angle (Xian et al., 2014), and it concentrates canopy rainfall on trunk roots to redistribute

rainfall into a pattern that is beneficial to vegetation growth (Wang C et al., 2019). In arid areas, for example, the stemflow of xerophytic shrubs increases the rainfall received per unit area of roots by 25–150 times compared with bare area (Li et al., 2008), and thereby provides sources of water for possible preferential flow into deeper soil layers, creating favorable soil water conditions for plant growth under arid conditions (Li et al., 2009).

Current theoretical and numerical modeling frameworks for canopy rainfall partitioning (including empirical, statistical, and physical mechanism models) are almost exclusively based on observations beneath woody plants, such as forests and shrublands (Gordon et al., 2020). However, few researchers have considered the temporal dynamics of canopy structure characteristics caused by plant growth (Wang G X et al., 2021), as well as the interception of herbaceous plants growing under forests and shrubs. As a result, the so-called net precipitation in many studies is not representative of the actual precipitation that reaches the soil surface (Gordon et al., 2020).

2.3.2 Soil water redistribution process induced by plant root

When the water potential in the soil is larger than that inside plant roots, which is often the case, water will flow into the roots—normally called “root water uptake”; when the water potential is opposite, water may flow from roots to soil—typically referred to as “hydraulic redistribution” (Nadezhdina et al., 2010). The two processes jointly control the soil water dynamics in the field profile during rainless days.

For a single plant, when plant transpiration demand is given, the root water uptake is a function of the rhizosphere water potential distribution, axial and radial root conductivities, and the three-dimensional architecture of the root system (Fatichi et al., 2016). That is, root water uptake depends both on the self-regulation of root conductivities and the spatiotemporal nature of soil water availability, which is the result of complex interactions between plants and soil (Doussan et al., 2006). Although our understanding of the root water uptake mechanisms has been improved over the past few decades, there remain many challenges—for example, the root conductivities can be partially controlled chemically by plant aquaporins, but the exact mechanisms underlying this control are still uncertain (Zhao et al., 2004). In addition, some studies have suggested that the occurrence of nighttime sap flow is beneficial for photosynthesis and transpiration the following morning and therefore helps shrubs adapt well to drought (Fang et al., 2017). An important limitation of the studies of root functioning and the development of models or conceptualization is the immense challenge of instrumenting and monitoring roots in field conditions (Fatichi et al., 2016; Yang et al., 2017), resulting in limited datasets of root water uptake processes (Dubbart

and Werner, 2019) and the fall-behind understanding of plant underground processes compared with the plant above-ground process.

Hydraulic redistribution, which is generally smaller in magnitude than root water uptake, widely exists in different ecosystems around the world (Caldwell et al., 1998), and it can be divided into hydraulic lift processes (i.e., plant roots transfer water from deep soil to shallow soil) and downward and horizontal hydraulic redistribution (top-bottom or horizontal transport) (Yang G S et al., 2021). Previous work shows that the moving water caused by hydraulic lift at night can account for 30–40% of daytime transpiration, which significantly promotes soil water redistribution (Horton and Hart, 1998). The controversies about hydraulic redistribution include (1) whether the phenomenon is a part of a typical plant life process or confined to specific conditions (Fatichi et al., 2016); (2) how much the phenomenon affects the global ecohydrological process (Yang G S et al., 2021); and (3) whether the hydraulic lift process is a positive feedback process—some researchers believe it can enhance nutrient cycling by wetting soil and using the nutrients absorbed to support root growth (Caldwell et al., 1998).

2.4 Interactions between vegetation patches and local hydrological processes (10^0 – 10^4 m)

2.4.1 Ecohydrological mechanism of vegetation patch formation

Self-organized spatial patterns of vegetation, which exist in forests, wetlands, grasslands, deserts, and other ecosystems (Li Y G et al., 2019), are the result of long-term interactions between soil, vegetation, and hydrological processes. In recent years, researchers have proposed many different hypotheses about the formation mechanism of vegetation patch patterns, based on field observations or modeling studies. Among them, the “scale-dependent feedback” theory is one of the most popular (Rietkerk and Van de Koppel, 2008); this theory suggests that the formation of vegetation patches is jointly controlled by short-distance positive feedback within patches and long-distance negative feedback between patches. In other words, the scale-dependent feedback of “soil-vegetation” and “vegetation-hydrology” leads to the formation of regular spatial patterns of ecosystems. For example, in some arid ecosystems, the higher the vegetation density is, the higher the soil infiltration rate, so patches can increase their amount of infiltrated water, which in turn promotes vegetation growth. Bare land easily forms soil biological crust, which increases runoff and reduces infiltration, limiting soil water replenishment and inhibiting vegetation growth (Zuo et al., 2021). Therefore, the positive feedback in the patch zone and negative feedback in the interpatch zone together promote the formation of vegetation patches in arid areas. The evidence of scale-dependent feedback for vege-

tation pattern formation is increasing rapidly (Table 1), and its physical process can be simulated by the so-called “activator-inhibitor principle” proposed by Turing, which is now a well-known universal principle explaining regular pattern formation in chemical and physical systems (Rietkerk and Van de Koppel, 2008).

Some previous studies suggested that the spontaneous formation of regular vegetation patterns can be regarded as an early warning signal for tipping points towards an alternative stable state. However, the latest findings indicate that such spatial self-organization should not necessarily be interpreted as an early warning signal for critical transitions. Instead, spatial self-organization can cause ecosystems to evade tipping points and thereby be a signal of resilience (Rietkerk et al., 2021). Future work should focus on better understanding and predicting regular pattern formation in ecosystems and how this affects the response of ecosystems to global environmental change.

2.4.2 Interactions between soil water erosion and vegetation patch patterns

Soil erosion resulting from water is a major issue in many areas, especially in drylands (Saco et al., 2018), and it is the result of the coaction of soil, vegetation, and hydrological processes. A large number of studies have shown that the vertical structure and horizontal distribution pattern of vegetation can have a significant impact on the process of soil water erosion. In terms of vertical structure, the aboveground traits of vegetation, such as vegetation type, canopy structure, coverage, and litter layer, can reduce the degree of water erosion by intercepting rainfall and weakening raindrop energy. The underground part of vegetation (root system) can reduce soil erodibility and surface runoff by improving soil structure and biophysical properties (Xu et al., 2006). Soil water erosion can also be affected by many factors except vegetation, such as terrain and rainfall characteristics, but in most cases, the higher the vegetation coverage proportion is, the weaker the soil water erosion (Gerwin et al., 2009; Hu et al., 2020; Cheng et al., 2021). Liu B Y et al. (2020) investigated soil erosion around China and found that intensive soil erosion occurred on cropland, overgrazing grassland, and sparsely forested land. Gao et al. (2021) investigated the shrub encroachment process in inner Mongolia, northern China, and found that the higher the shrub coverage was, the higher the soil water content, field capacity, soil clay, and organic matter content.

The horizontal distribution pattern of vegetation patches can change the soil water erosion process by affecting the source-sink spatial distribution of runoff and sediment and their connectivity (Imeson and Prinsen, 2004). For example, in arid areas, striped vegetation patterns orthogonal to the slope surface can effectively limit soil erosion through runoff generation and accumulation systems (Xu et al., 2006). Saco

Table 1 Scale-dependent feedback of vegetation patches in different ecosystems

Example	Feedback mechanism
Feedback of restrictive nutrients in the wetland ecosystem	Nutrients in groundwater are convectively transported to high plant biomass areas due to the differences in transpiration rates. Therefore, high vascular plants enhance their competitiveness by affecting hydrological processes (Rietkerk et al., 2004).
Feedback of local nutrient accumulation in the savanna ecosystem	Tree litter and animal excrement under trees fertilize the nutrient-poor soil, which has positive effects on plants themselves (Belsky, 1994).
Snow feedback of alpine forest ecosystem	A large amount of snow accumulates between tree patches, which limits the establishment of tree's seedlings (Rietkerk and van de Koppel, 2008).
Temperature-microorganism feedback in Arctic tundra-shrub ecosystem	The shrubs intercept more snow in winter, which increases soil temperature, resulting in more microbial activity and produces more available nitrogen for shrubs growth in the coming summer (Sturm et al., 2005).
Shading feedback in an arid ecosystem	The soil evaporation in the patch shadow area is lower, which can preserve soil water and is conducive to vegetation growth (Liu Q S, 2020).
Soil water diffusion feedback in an arid ecosystem	The water consumption caused by plant roots promotes soil water around patches to move into patches (Liu Q S, 2020).

et al. (2018) reported that dryland vegetation is organized in patterns for improved water capture, and these patterns emerge from nonlinear soil-vegetation-hydrology feedbacks. Ludwig's study of vegetation patches in global semiarid areas shows that the reduction in vegetation coverage induced by environmental perturbations can significantly increase runoff generation and sediment loss during rainstorms, triggering erosion and desertification (Ludwig et al., 2005).

In turn, soil water erosion also impacts vegetation patches by causing soil particle and nutrient losses (McIvor et al., 1995), reducing the water holding capacity, and affecting plant seed survival (Espigares et al., 2011). In fact, the vegetation patch pattern and soil water erosion process are interactive and codeveloped (Puigdefàbregas, 2005), which reflects the theory of "pattern controls the process, process influences pattern" in landscape ecology. Some modeling studies show that both the bare land and homogeneous planted land in the arid area will eventually evolve into a striped vegetation pattern, because this is the best spatial pattern for vegetation growth under specific climate, terrain and soil conditions—it can limit land degradation while increasing biomass (Xu et al., 2006).

The knowledge gaps in this research field include the following: (1) lack of a large-scale soil water erosion model that can adapt to complex environments. At present, observation and experimental research on the soil water erosion process have mostly focused on small space-time scales, which limits the development of mechanism models for large-scale soil erosion (Xu et al., 2006; Wang H et al., 2021). (2) Few studies have focused on the adaptation mechanism of plants to soil erosion stress. Research on the erosion resistance mechanism of erosion-resistant plants is of great significance to the practice of vegetation restoration (Guerrero-Campo and Montserrat-Martí, 2000; Jiao et al., 2012). (3) The existing vegetation patch pattern index mainly focuses on the simple description of the pattern and lacks an index that can reflect the ecological process. The special

spatial arrangement of vegetation patches can produce edge effects, causing (air) advection and circulation and affecting ecological processes (Anderson et al., 2003), which are research hotspots in landscape ecology (Xu et al., 2006).

2.4.3 Influence of vegetation patches on local hydrological processes

Compared with leaves and individual plants, the behavior of vegetation patches is more complex. An increase in spatial scale amplifies the importance of the feedback process between vegetation and atmosphere and the horizontal hydrological connection between vegetation patches. Previous studies often focus on the impact of precipitation on vegetation; in fact, vegetation can also affect precipitation by affecting land-atmosphere energy and water exchanges. The existence of vegetation patches enhances the horizontal gradient of surface sensible and latent heat (Fatichi et al., 2016); some investigations suggested that landscape discontinuities, such as irrigated land in arid areas, deforestation, and afforestation, can induce atmospheric mesoscale circulations and have a strong impact on clouds, and precipitation (Avisar, 1998). For example, a survey found that the deforestation of the Amazon rainforest changed the downwind convective precipitation of felled forest patches (Knox et al., 2010), and similar phenomena of "vegetation affect precipitation" have been widely reported (Scheffer et al., 2005; Zhang B et al., 2021). Compared with bare land, vegetation patches can absorb more radiation and transpire more water to the atmosphere, causing a higher precipitation probability (Dekker et al., 2007). In addition, many tree species can release isoprene gas, whose oxidation products can form aerosols that can induce rainfall formation (Taylor et al., 2018). Zhang et al. (2001) studied more than 250 watersheds in the world and found that for a given forest cover, there is a good relationship between long-term average evapotranspiration (related to vegetation) and long-term average rainfall.

At the same time, the exist of vegetation patches can lar-

gely affect the infiltration processes through preferential flow. Previous studies showed that the formation of macropores due to root decay of mature plants often increase infiltration rate (Mitchell et al., 1995; Wang C et al., 2020). Root-induced preferential flow may be more significant when the root length density and biomass are high (Zhang et al., 2015; Shao et al., 2017). The velocities of observed macropore flow were about 100 to 400 times faster than the flow within the soil matrix (Bronstert and Plate, 1997), significantly changing the infiltration processes of one soil layer or even the whole soil profile. When transition to the patch scale, the features observed at the individual scale (i.e., hydraulic nonequilibrium and preferential flow) persist (Vogel, 2019). Besides, topography significantly controls surface and subsurface flow processes, such as preferential flow, at various scales (Frei and Fleckenstein, 2014). Many investigations have observed lateral preferential flow in the hillslope hydrological processes; however, conceptualizing and parameterizing the effects of lateral preferential flow is still a great challenge in the field of hillslope hydrology (Weiler and McDonnell, 2007). Representation of microtopography in numerical models usually requires high grid resolutions to capture relevant small-scale variations in topography in the range of centimeters to meters, which usually results in longer simulation times (Frei and Fleckenstein, 2014). The impact of vegetation patch patterns on horizontal hydrological processes has been discussed in many studies, and the most common case is the “patch-runoff interaction” in arid ecosystems: vegetation patches can intercept surface runoff as well as the sediments and nutrients brought by runoff; then, the increased water and nutrient availability of patch soil can promote vegetation growth, leading to more intercepted runoff (Ludwig et al., 2005). The spatial pattern of vegetation patches is often integrated into hillslope-scale ecohydrological models to explain the impact of vegetation patterns on horizontal hydrological processes (Saco et al., 2007). For example, Hwang et al. (2009) established a hillslope ecohydrological model considering lateral hydrological connectivity, and the results suggested that less upslope vegetation produces a subsidy of more water to downslope vegetation, where more water and nitrogen are available; therefore, the existing vegetation pattern must be understood as feedback between multiple stresses (e.g., light, water, and nutrients) connected by water flow along topographic gradients. Compared with homogeneous ecosystems, patches with specific arrangements may have a higher anti-interference ability to environmental changes (Rietkerk and van de Koppel, 2008).

However, most existing ecohydrological models only consider the vertical water movement (one-dimensional) that relates to vegetation patches (Gao et al., 2018), while the horizontal hydrological process between patches (two-dimensional) and the carbon, nitrogen and nutrient element

exchange brought by this process have not yet been fully considered (Hwang et al., 2009); in particular, the vegetation dynamics were largely underestimated in these ecohydrological models. Meanwhile, the vegetation-precipitation feedback mechanism is not explicitly represented in most land surface models (LSM) or Earth system models (ESM), even though a lack of relevant feedback mechanisms can severely impair the model hypothesis and incorrectly estimate the response of vegetation to environmental change (Dekker et al., 2007).

2.5 Links between plant community succession and soil development (10^2 – 10^4 m)

2.5.1 Mechanisms of plant-soil feedback

As plants grow, they modify the surrounding soil environment, which may in turn change the plant's physiological activity and/or plant population—that is, the outcome of the process causes the process itself to increase (positive feedback) or decrease (negative feedback) in magnitude—this is usually called plant-soil feedback (PSF) (Ehrenfeld et al., 2005). The primary PSF mechanism includes nutrient-mediated feedback, mutualist-mediated feedback, and secondary chemistry-mediated feedback (Bennett and Klironomos, 2019). For example, litter is the main source of soil nutrients and an important part of plant nutrient cycling (Giweta, 2020). For faster growing species, high-quality litter that decomposes rapidly can improve soil nutrient availability and promote plant growth. Fast-growing plants can produce more high-quality litter, initiating positive PSF. For slow-growing species with lower nutrient requirements, the nutrient depletion caused by low-quality litter can help them gain a competitive advantage over faster growing species (Bennett and Klironomos, 2019). Some studies show that invasive plants can stimulate nutrient release through the mechanism of litter feedback and thus promote their own invasion (Zhang P et al., 2019). Mutualist-mediated feedback refers to the interaction between plant roots and soil fungi; this effect usually produces positive feedback and can offset the negative effects caused by nutritional constraints. Secondary chemicals are non-nutritive substances produced in organisms. They can inhibit the growth of same-species (or cross-species) vegetation and their mutualists, resulting in negative or positive PSF (Bennett and Klironomos, 2019).

At present, plant-soil feedback mechanisms have not been fully investigated. For example, different PSF mechanisms are usually studied separately, and the contribution of each PSF mechanism and their interactions are rarely quantified (Van der Putten et al., 2016; Bennett and Klironomos, 2019). Whether the PSF mechanism obtained from indoor experiments is consistent with that under natural conditions is debatable (Beals et al., 2020). The methods of studying agricultural system PSF and natural system PSF are largely

independent (Mariotte et al., 2018). Finally, there is a lack of methods for integrating the PSF mechanism at different spatial scales.

2.5.2 Interactions between soil development and plant community succession processes

As one of the main driving factors for soil development, plants can affect soil physical and chemical properties by litter accumulation, root growth, root exudate import, dust interception, changing soil microorganisms, and weathering bedrock over a longer time scale (Paterson et al., 2007; Su and Liu, 2020). Many studies have shown that the establishment and succession of different plant communities can lead to significant changes in the soil-forming rate and then affect soil hydraulic characteristics and, therefore, the water balance (Gutiérrez-Jurado et al., 2006). For example, Li Z K et al. (2019) showed that in oasis areas, after cultivating different crops in fields with initially the same soil texture for 10 years, the Ks and soil water retention curve under different crops become different, resulting in different field hydrological processes. In more complex terrains, such as hillslopes, the influence of the plant community on the soil forming process is more complex. Hillslope plants can affect soil development by affecting the production, transportation, deposition, and erosion of soil materials (Gutiérrez-Jurado et al., 2006). However, corresponding studies are quite limited.

On the other hand, in some cases, soil characteristics control the establishment and succession of plant communities. For example, Hahm et al. (2019) studied two adjacent small watersheds in California and showed that the difference between the bedrock lithology and slope gradient resulted in different weathering degrees in the critical zones, which affected the soil thickness and thus the water storage capacity, eventually causing two contrasting plant communities under the same climate. Henne et al. (2011) showed that in the Alps, spruce (*Picea abies*) would expand in the Central Alps only when the soil develops high water holding capacity. In conclusion, the degree of soil development controls the establishment of plant communities by soil thickness, availability of soil water and nutrients, activities, and types of soil microbes (Fernandez-Illescas et al., 2001; Chai and Yue, 2016; Cui et al., 2020). Changes in soil characteristics (caused by factors such as soil erosion) can induce plant community succession by affecting the yield, redistribution, and survival rate of plant seeds (Espigares et al., 2011).

Numerous studies have shown that there are significant interactions between plant community composition and soil properties (Maxwell et al., 2018). One of the research weaknesses in this field is that the role of the terrain is not fully understood in the interaction process of soil development and plant community succession, and knowledge about the controlling mechanism of terrain on hillslope ecopedo-

logical processes is limited (Gutiérrez-Jurado et al., 2006). In addition, the impact of soil nutrient limitation (especially phosphorus limitation) on plant and soil microbial community succession is not totally clear (Cui et al., 2020).

2.6 Links between watershed/regional water budget and vegetation phenology and production (10^4 – 10^5 m)

2.6.1 Hydrological factors controlling vegetation distribution

The global distribution of vegetation types is strongly controlled by precipitation (Stephenson, 1990). More specifically, the soil water storage in the root zone, input from precipitation, directly determines the structure, type, and function of ecosystems (Ludwig et al., 2005; Maestre et al., 2012; Hahm et al., 2019). For example, Singh et al. (2020) studied six typical vegetation transects in South America and Africa and showed that the main reason for the transformation of different vegetation types was the change of water storage capacity in the root zone, and suggested that the root zone water storage capacity is a good measure in indicating the transition behavior between different ecosystems; Tian et al. (2019) obtained the root-zone water storage of global vegetation using data assimilation method and found that it has a high correlation with global NDVI; Liu L B et al. (2020) quantified the relative effect of global surface soil water and VPD in limiting ecosystem productivity using remote sensing data and found that more than 70% of global vegetation drought stress is dominated by soil water rather than VPD. Zhou et al. (2019b) showed that the compound water stress of low soil moisture and high VPD would lead to stronger declines in vegetation production using 15 ESM. In addition, in some arid areas, groundwater level fluctuations also significantly change vegetation patterns (Cooper et al., 2006). Roebroek et al. (2020) analyzed the relationship between global atmospheric humidity, groundwater depth, and stand development and found that vegetation and groundwater depth show different degrees of correlation at the subclimate scale.

A limited number of case studies have shown that plant water uptake is not restricted to shallow soils but can involve uptake from rock layers below (Schwinning, 2010). Some studies have suggested that different bedrock features can translate into contrasting belowground water availability, leading to landscape-level heterogeneity of tree water status and drought response (Nardini et al., 2021). Research on the mechanisms governing root-rock interactions is still in the early stage (Schwinning, 2010). Many questions need to be solved: (1) how much water do plants draw from rocks, and under what climatic circumstances? (2) What are the pathways by which water moves from rock to the root surface? (3) How should the rock fraction of plant-available water capacity (PAWC) be represented in models (Schwinning,

2020)? These questions need to be further explored in order to develop a better understanding of the coupling mechanism of plant and bedrock hydrological processes.

2.6.2 Impact of vegetation changes on watershed/regional water budget

In recent years, the phenomenon of “global vegetation greening” has attracted widespread interest (Piao et al., 2020); as a result, vegetation greening-induced changes in regional hydrological processes and water resources have gradually become a research hotspot in hydrology and ecology (Zhou et al., 2021). In earlier studies, the impact of vegetation changes on catchment evapotranspiration and runoff was primarily assessed through paired catchment studies. However, the relationship between vegetation change and hydrological processes is highly nonlinear; it is strongly influenced by climate and catchment characteristics (topography, soil type, vegetation type, etc.) (Zhang and Wei, 2021). The research results of paired catchment studies in one region cannot be simply popularized in other regions (Peel et al., 2010). Taking runoff as an example, in different paired catchment experiments, an increase in vegetation can increase, decrease or not affect catchment runoff. Zhang et al. (2017) examined approximately 312 watersheds worldwide to evaluate hydrological responses to forest cover change, and found that the increase in annual runoff associated with forest cover loss is statistically significant at multiple spatial scales, whereas an increase in forest cover on runoff is statistically inconsistent; at the same time, the sensitivity of annual runoff to forest cover change tends to attenuate as watershed size increases in large watersheds ($>1000 \text{ km}^2$). Zhou et al. (2021) used Fu's equation to deduce and analyze the relationship between vegetation change and annual water yield and verified the result with the long-term monitoring data of 1928 watersheds worldwide; they found that in watersheds with small areas and low water retention capacity ($m < 2$), an increase in vegetation cover can reduce the water yield of the watershed by improving the water retention capacity, but in some complex watersheds with large areas and high water retention capacity ($m > 2$), an increase in vegetation will not change the runoff. Luo et al. (2020) quantified the impact of vegetation change on surface runoff based on the Budyko framework from 663 strictly selected catchments globally during 1982–2014; they found that vegetation greening-induced runoff declines were manifested in global drylands but negligible in humid areas.

In addition to runoff, numerous studies have been performed to investigate the impact of vegetation change on watershed evapotranspiration, soil water, and even regional precipitation. Early studies suggested that vegetation greening increased evapotranspiration and reduced runoff and soil moisture in some regions (Yang et al., 2010). For example, some studies show that the “Grain-to-Green Pro-

gram” project in the Loess Plateau has significantly increased vegetation evaporation (water consumption) in this area (Shao et al., 2019) and reduced streamflow, resulting in soil drying in some subareas (Feng et al., 2016), and the soil drying depth caused by artificial afforestation is deeper than that caused by natural revegetation (grassland) (Ren et al., 2018). Peel et al. (2010) analyzed the impact of vegetation types on annual average evapotranspiration in 699 watersheds around the world and found that this impact will decrease with the increasing watershed area. Piao et al. (2020) believe that “global vegetation greening” enhances transpiration, which reduces soil water and runoff in some areas, but in other areas, the increase in transpiration and thereby moisture recycling for precipitation increases soil water and runoff. Li Y et al. (2018) investigated 30-year vegetation impacts on regional precipitation using a coupled land-atmosphere global climate model (IPSLCM GCM), and found that in north and southeast China, the increased precipitation caused by vegetation greening is of a similar amount as evapotranspiration increases, resulting in a weak impact on soil moisture; in southwest China, however, the increase in vegetation cover significantly reduced soil moisture, and the precipitation was suppressed by the weakened summer monsoon. Zeng et al. (2018) used similar methods to study the impact of global vegetation dynamics on the water budget and found that in humid areas, “greening” increased evapotranspiration and precipitation at the same time, so it did not significantly reduce runoff and soil moisture; however, in arid areas, “greening” significantly reduced soil moisture.

The impact of regional vegetation change on groundwater is slightly complex. On the one hand, vegetation increases enhance soil infiltration, which may be conducive to groundwater recharge. On the other hand, vegetation can reduce groundwater recharge by consuming soil water (Yang et al., 2010). Previous studies have shown that an increase in vegetation increases the fluctuation range of groundwater (Martinet et al., 2009); at the same time, human activities may have a greater impact on groundwater. For example, in some cases, the fluctuation of groundwater level caused by oasis agricultural irrigation is significantly greater than natural groundwater fluctuation in riparian zones (Wang et al., 2014). Vegetation change is also closely related to canopy interception. On a global scale, the higher the vegetation coverage and annual rainfall, the higher the total canopy interception (Zheng and Jia, 2020). In addition, plant phenological changes can also affect hydrological processes. Guo et al. (2020) showed that the start of the growing season is highly correlated to the evapotranspiration of subtropical forests, in Zhejiang Province, China. Geng et al. (2020) revealed that the advanced start of the growing season and delayed end of the growing season induced an extension of growing length and thus caused increased evapotranspiration

and decreased runoff at the basin scale. [Lian et al. \(2020\)](#) found that the advancement of vegetation phenology in the Northern Hemisphere increased spring evapotranspiration and thus caused a soil water deficit in the summer.

Overall, current researches lack large-scale, long-term, consistent, and comparable in-situ observations to mechanistically study and analysis the effect of large-scale re-vegetation projects on local hydrological processes. Meanwhile, the verification of watershed-scales simulation is usually constrained by the lack of observation data. Besides, only few models have addressed the processes of temporal variation of soil properties ([Lu et al., 2020](#)) and plant traits and feedbacks of vegetation-hydrology, when simulating the large-scale, long-term watershed water budget.

In addition, under the background of climate change, researchers should pay more attention to the impact of special vegetation changes—such as shrub encroachment, alien species invasion, forest death, and artificial afforestation—on ecohydrological processes. These special vegetation changes are directly related to the safety of water, food and industrial materials resources ([Zhang, 2018](#)), and researches in this field can not only make theoretical advances in ecology and hydrology, but also provide a scientific basis for realizing the sustainable development of human society ([Liu et al., 2021](#)).

2.7 Impacts of climate change and human activity on coupling mechanism of soil-vegetation-hydrological processes

2.7.1 Impacts of climate change on coupling mechanism of soil-vegetation-hydrological processes

Rising atmospheric CO₂ and associated climate changes in temperature and precipitation can impact the strength and direction of coupled soil-vegetation-hydrological processes, by impacting plant physiology properties and production, soil microbial activity, soil moisture, runoff, soil erosion intensity, and so on ([Li and Fang, 2016](#), [Silva and Lambers, 2020](#)). The thermodynamic consequence of climate warming can lead to the increase of VPD and potential evapotranspiration (PET) as well as the decrease of soil moisture, which further amplify atmospheric drying ([Zhou et al., 2019a](#)), while the increase of CO₂ concentration can reduce stomatal conductance and transpiration water loss, at the same time increase WUE ([Lian et al., 2021](#)). During this process, the coupling mechanisms of soil-vegetation-hydrological processes—such as plant-soil feedback, vegetation-precipitation feedback, and canopy rainfall partitioning—will be affected to some extent ([Nie et al., 2013](#); [Li Y et al., 2018](#); [Xia et al., 2020](#); [Lian et al., 2021](#)).

For example, [Harrison et al. \(2014\)](#) found that in some areas with infertile soils, elevated temperature and CO₂ can

increase the competitive advantage of nonmycorrhizal plants. [Silva et al. \(2016\)](#) showed that, at Tibetan Plateau, tree growth had been stimulated by the synergistic effect of rising atmospheric CO₂ and a warming-induced increase in water and nutrient availability from thawing permafrost. [Maxwell et al. \(2018\)](#) showed that, in California, climate change have caused alterations in forest species composition, iWUE and productivity. Some studies suggested that changes in precipitation patterns may affect the soil-vegetation relationship: plants can exist preferentially on coarse soils under insufficient rainfall conditions, however, when rainfall is sufficient, they may exist preferentially on finer soils ([Noy-Meir, 1973](#); [Fernandez-Illescas et al., 2001](#); [Wang C et al., 2019](#)).

At the same time, the availability of soil water and nutrient resources may partly control the impact of climate change on soil-vegetation-hydrological processes coupling mechanism. For example, as mentioned above, for watersheds with different water retention capacity, the response of the hydrological process to “vegetation greening”—which is mainly driven by elevated CO₂, according to [Piao et al. \(2020\)](#)—is quite different. Much evidence suggests that plant communities on infertile soils are relatively insensitive to increased water deficit caused by climate change ([Harrison et al., 2014](#)). [Rowland et al. \(2021\)](#) investigated tropical forests around the world and found that in well-watered trees, their growth is controlled by traits related to carbon cycling (assimilation and respiration), while the growth of drought-stressed trees is controlled by traits related to hydraulic stress. Other studies showed that rapid (days to weeks) short- and long-distance (centi-meters to tens of meters) transfer of nutrient resources could occur in root-mycorrhizal networks as plant and microbial communities respond in climate warming ([Bingham and Simard, 2012](#); [Silva and Lambers, 2020](#)). However, the direction of this transfer is controlled by the availability of soil resources ([Silva and Lambers, 2020](#)).

Future work about soil-vegetation-hydrology coupling must consider the impact of climate warming and elevated CO₂ ([Wang C et al., 2019](#)), and the degree of this impact is controlled by the availability of soil water and nutrient resources, as well as the trade-off of root-mycorrhizal carbon/water budget ([Silva and Lambers, 2020](#); [Lian et al., 2021](#)). At present, uncertainties are still involved with respect to specific influencing mechanisms and actual damages, and the impacts vary with different regions ([Li and Fang, 2016](#)).

2.7.2 Impacts of human activity on coupling mechanism of soil-vegetation-hydrological processes

Earth is now in the Anthropocene era, in some instances, human activities are playing a central or even dominant role in shaping terrestrial vegetation patterns, modifying regional hydrological cycles, and changing soil characteristics ([Kenchington, 1943](#); [Tang, 2020](#); [Lian et al., 2021](#)). More

specifically, the coupling mechanism of soil-vegetation-hydrological processes is affected by both direct human activities, such as vegetation restoration, overgrazing, dam construction, intense irrigation, excessive groundwater pumping, soil pollution and land-use changes (Wada et al., 2016; Li X et al., 2018a; Chen C et al., 2021; Shah et al., 2021), and indirect human forcing such as human-induced CO₂ and temperatures rise.

The impacts of human activities on the regional hydrological cycles have been widely studied through different methods, such as paired catchment method, modeling method, Budyko-based approach, and time trend approach. For example, Brooks et al. (2003) studied a paired catchment in Australia, and found that Thurra River is effectively in the same hydrological condition today as it was at 150 years ago; in contrast, the adjacent Cann River, which has been settled by Europeans, exhibits stark differences today. Limitations of the paired-catchment approach are that finding a suitable catchment pair can be very challenging, and long time series of hydrological data are needed to robustly diagnose the effect of human activities on hydrological processes (Van Loon et al., 2019). Most recent studies quantifying the human influence on hydrological processes use hydrological models because of their ability to isolate processes by generating scenarios with and without human activities for comparison (Wada et al., 2016), and the results are usually different from place to place. For example, Yang et al. (2020) investigated how human activities have altered hydrological droughts (streamflow deficits) in China during 1961 to 2016, using PCR-GLOBWB model, and found that over northern China, human activities generally intensify hydrological droughts, while droughts over southern China are generally alleviated by human activities. Shah et al. (2021) used the Community Water Model (CWatM) to quantify the influence of human activities on terrestrial water fluxes in India from 1951 to 2016. The results showed that evapotranspiration and soil moisture increase while total runoff and groundwater storage decrease under the various agricultural activities (e.g., irrigation, reservoir storage, and groundwater pumping). One of the main limitations in modeling studies could be the uncertainty associated with inputs, model assumptions, and parameterization. Besides, there are still some known anthropogenic processes that influence hydrological processes that have not been included in hydrological models (Van Loon et al., 2019). The Budyko-based method and the time-trend method are also very common in quantifying and isolating the hydrological impacts of human activities, due to the simplicity and the few data requirements. For example, Wu et al. (2017) used a technique involving integrated climate elasticity and eight Budyko-based methods to assess the relative impacts of climate change and human activities on runoff changes in 17 catchments across the Loess Plateau during the period 1961–2013, and found

that human activity contributed on average 73% of the decrease in the high-flow season. Zhang L et al. (2019) distinguished the contributions of climate change and human activities to the variations in streamflow, baseflow, and surface runoff in the upper Heihe River Basin by using eight different forms of time-trend methods over the period 1960–2014. However, most of the Budyko-based methods and the time-trend methods have not considered the hydrological impacts of the changes in cryospheric processes resulting from climate variability, which may lead to underestimations or overestimations of the contributions of climate changes and human activities. On the other hand, human activities also have significantly altered the vegetation patterns and soil characteristics, especially in dryland. Over the last 100 years, global crop land area has doubled to 15 million km², while the global irrigated area has increased six-fold from 0.5 million km² to 3.0 million km² (Wada et al., 2016). In India and China, agricultural expansion and afforested activities have evidently increased regional greening and vegetation productivity (Chen et al., 2019; Piao et al., 2020). Agricultural expansion and intensification are often accompanied with excessive withdrawal of surface waters (mainly runoff) and overextraction of groundwater storage (Lian et al., 2021), which in turn affect the health of natural species that depend on shallow groundwater for survival, particularly during dry seasons (Wang et al., 2014). In other cases, the implementation of effective basin management brought about improved environment conditions, with a more optimal hierarchical structure of vegetation cover (Zhou et al., 2015). Meanwhile, continuous cultivation will have a significant impact on soil physical, hydraulic and biological properties. For example, a large number of researches have shown that long-term cultivation and tillage management could significantly affect the soil organic carbon content, bulk density, saturated hydraulic conductivity, water retention, and pore-size distributions in a different type of soils (Mudgal et al., 2010; Page et al., 2020). Other research showed that the soil microbial community and nutrient availability in the cultivated land varied from it in native lands (Zhang Y et al., 2016; Ashworth et al., 2020).

Overall, human activities have both positive and negative effects on the coupling mechanism of soil-vegetation-hydrological processes (Li X et al., 2018a), depending on the climate, economic, population, and management policies of the study area. Future work should strengthen the availability of high-resolution data on human activities, and take more anthropogenic processes into account, to provide guidance for more effective basin management.

2.8 Representative studies for coupled soil-vegetation-hydrological processes in Heihe River Basin

A watershed is a basic unit and possesses all of the com-

plexities of the land surface system (Cheng and Li, 2015; He and James, 2021), thereby making it the best unit for practicing Earth system science and studying the coupling mechanism of soil-vegetation-hydrological processes. Over the past ten years, several watershed observation systems with advanced observation techniques have become well established, such as the Danish Hydrological Observatory (HOBE), which was established in 2007 in a catchment in the western part of Denmark (Jensen and Illangasekare, 2011); and the watershed ecohydrological observation system (HiWATER) established through the “Integrated research on the eco-hydrological process of the Heihe River Basin” project supported by the National Science Foundation China since 2010 (hereinafter referred to as the “Heihe Plan”).

The Heihe Plan is by far the largest watershed research project in China. Since its implementation, scientists from different disciplines have made a series of progress in multi-scale observations, scaling methods, mechanisms and modeling approaches of coupled soil-vegetation-hydrological processes (Song, 2019). Over the past few decades, the number of published articles that relate to the Heihe River Basin (HRB) has ranked in the top 20 in the global watershed science research, and more than 14 articles supported by the Heihe Plan are in the top 1% cited papers in the basic science indicators (ESI) (Wang X M et al., 2019).

For example, a world’s leading multi-factor, multi-process, multi-scale watershed integrated observation system—HiWATER—has been well established, through instrumentation on hierarchically nested scales—from point to pixels, sub-basins, and the whole river basin—to obtain multiscale and multidisciplinary data and address the challenge of upscaling of eco-hydrological processes on a heterogeneous surface (Li et al., 2017). Based on these intensive, three-dimensional, and advanced observations (Cheng et al., 2020), several upscaling methods for soil moisture, evapotranspiration, and LAI are developed (Li X et al., 2016), using theory frameworks such as regression Kriging, Bayesian maximum entropy, measure theory and stochastic calculus (Li, 2014; Li X et al., 2016; Liu and Li, 2017).

A series of scientific questions in the coupling of soil, vegetation and hydrological processes have also been explored through Heihe Plan. For example, Li X et al. (2018a) analyzed the hydrological cycle in the HRB using both ecohydrological model and systematic observation data and suggested that the natural oasis ecosystems were considerably restored with the groundwater table rose in the downstream area. He Z B et al. (2012a) examined the effects of different vegetation on hydrology in the upstream area, and found that grassland contributed the largest proportion of annual water yield (between 53 and 60%), followed by bare rock and alpine meadow. Besides, increased forests area may decrease the downstream water supply. His other research

showed that the responses of soil moisture in grassland and meadow to rain event size were similar, although the rainfall characteristic and vegetation types were different (He Z B et al., 2012b). Li and Shao (2013) investigated soil-profile textures at 100 sites in the central region of the HRB, and characterized the vertical transition of the soil textural layers by a Markov chain-log-normal distribution (MC-LN) model, aiding in the quantification of water transport in soil. Other studies in the HRB showed that the soil organic matter and porosity in forest land are larger than that in shrubs, leading to a difference in soil water capacity (Cheng et al., 2020). Yu et al. (2013) studied the hydraulic redistribution processes of *P. euphratica* and *T. ramosissima*, and showed that different plants exhibit remarkable differences in their patterns and pathways of hydraulic redistribution, which are possibly associated with root architecture. Li W et al. (2016) studied the hysteresis between trunk sap flow velocity and canopy transpiration in desert riparian forests in the HRB, and found that hysteresis was primarily caused by the biophysical declining in canopy conductance, which is affected by different environmental variable during different time. Zhou et al. (2018) used two underlying WUE (uWUE) formulations to estimate T/ET in the HRB; this method can estimate ecosystem T/ET without scaling issues, and can effectively capture T/ET variations in relation to LAI changes.

Many models have also been developed during the implementation of Heihe Plan. For example, two integrated ecohydrological models, i.e., the geomorphology-based ecohydrological model (GBEHM) (Yang et al., 2015) and the hydrological-ecological integrated watershed-scale model (HEIFLOW) (Tian et al., 2018), were developed during the past several years (Li et al., 2021). HEIFLOW is mainly composed of surface water model PRMs, groundwater model MODFLOW and special ecological modules developed according to the soil, vegetation and climate characteristics of the HRB (such as desert vegetation module, dynamic land-use module, irrigation module and canal system leakage module). GBEHM is constructed by coupling modified GBHM model with cryosphere hydrological process and dynamic vegetation process, and it therefore can accurately simulate the ecological hydrological process in Alpine mountainous areas (Cheng et al., 2014, 2020). Besides, a watershed system model (WSM) that couples the integrated ecohydrological model with the socioeconomic model was developed for the HRB (Li et al., 2021). This model showed good abilities in exploring the trajectories and interactions of water, ecology and socioeconomy at the river basin scale, and has great implications for sustainable developments in endorheic river basins (Li X et al., 2018b). Regarding the significant uncertainty for modeling at a large basin scale, Wu et al. (2014) presented an innovative framework of conducting systematic uncertainty analysis, using Probabilistic Collocation method, for integrated surface water-

groundwater modeling of HRB.

3. Spatial upscaling of coupled soil-vegetation-hydrological processes

Existing studies on the interaction of soil, vegetation and hydrological processes mostly focus on small temporal and spatial scales (0.01–10 m², month-year) (Vogel, 2019; Silva and Lambers, 2020). However, policymakers usually make water resource management policies on a watershed, regional or global scale. Therefore, how to build bridges between different scales and how to predict large-scale phenomena through small-scale processes is a major challenge for scientists. In modern geography, the method of upscaling can be divided into two categories: (1) if the mechanism of small scale is still applicable on a larger scale, upscaling can be carried out only by changing the resolution of the model; (2) if the mechanism of small scale is not applicable on a larger scale or needs to be modified before it can be applied, it is necessary to establish the connection between mechanisms at different scales (Zhao et al., 2002). This article focuses on the latter category and reviews the upscaling method of “coupled vegetation-hydrological processes” and “soil hydrological process” to show the spatial upscaling of coupled soil-vegetation-hydrological processes.

3.1 Spatial upscaling of coupled vegetation-hydrological processes

The upscaling method of the “coupled vegetation-hydrological process” can be summarized into three types, namely, empirical statistics method, mathematical model method, and parameter equivalent method. The metabolic theory of ecology provides a biological theory basis for the cross-scale connection between plant tissue, individual plants and ecosystems (Brown et al., 2004). The metabolic theory holds that there is a power relationship between the plant metabolic rate and individual size, and this relationship can be extended from the plant tissue scale to the ecosystem scale; at the same time, the metabolic rate is directly related to vegetation hydraulic characteristics (Wang et al., 2009; Guo, 2011; Liu and Xia, 2011). For example, plant growth strategies can be summarized using leaf morphology; thus, leaf morphology can predict stomatal behavior effectively (Kröber et al., 2015). The relationship of “leaf morphology-stomatal behavior” can be extended to the landscape scale using SLA (specific leaf area) and LAI (leaf area index) data, which allows the leaf-to-canopy upscaling of photosynthesis and transpiration (Sprintsint et al., 2012). Finally, the biological forces that drive the mass flux at the canopy scale also operate at the Earth system scale (Silva and Lambers, 2020).

In empirical statistical methods, upscaling can be im-

plemented by establishing the statistical relationships among tissue-scale functional traits, individual-scale traits, and ecosystem characteristics. For example, Christoffersen et al. (2016) improved the performance of a forest model using an empirical statistical relationship of hydraulic properties between vegetation tissues (leaves and xylem) and individual plants. Taking the upscaling study of water use efficiency as an example, a large number of empirical data (or statistical relationships) have shown that if VPD remains unchanged, iWUE (leaf photosynthetic/stomatal conductance, stomatal scale) and WUE (biomass/transpiration, plant scale) are linearly correlated. Therefore, iWUE and WUE can be connected through VPD (Beer et al., 2009), that is, $iWUE = WUE \times VPD$. Through this connection, leaf-to-plant upscaling of water use efficiency can be implemented. VPD is relatively stable within the ecosystem, so leaf-to-ecosystem upscaling of water use efficiency can be implemented by using specific leaf area (SLA) data and standardized species iWUE (Zhou, 2017; Silva and Lambers, 2020). It should be noted that spatial upscaling and temporal upscaling can be completed simultaneously by using isotope means: iWUE is generally measured through the $\delta^{13}C$ isotope ratio in plant cellulose; meanwhile, the ^{13}C isotope value of soil organic matter can be used to estimate the stand-level LAI (Ladd et al., 2014), and this value can be used to calculate canopy density. Therefore, a combination of plant cellulose ^{13}C and soil organic matter ^{13}C can accomplish the spatial upscaling between ecosystem-scale WUE (i.e., GPP/ET) and iWUE. At the same time, the isotope value of tree rings provides a means to reconstruct iWUE in historical periods (Dietrich et al., 2016) (i.e., temporal upscaling). Therefore, by combining isotope measurements and remote sensing data, upscaling of water use efficiency across time and space can be completed simultaneously (Correa-Díaz et al., 2019).

The mathematical model method mainly includes a stochastic ecohydrological model and capacitance-resistance model. Mencuccini et al. (2019) summarized the upscaling steps of most stochastic ecohydrological models: (1) relations between stomatal conductance or xylem conductivity and leaf water potential are scaled up to the stand level using leaf and sapwood area indices (LAI and SAI); (2) stand level relations between transpiration rate (E) or net primary productivity (A) and soil moisture (S) obtained from the scaled-up properties from step one; and (3) the probability density function of S is obtained by random rainfall events, and then the probability density functions of E and A are obtained from the results of step two. Because plant traits determine the shape of transpiration and productivity response functions, the probability distribution functions of water and carbon fluxes in these models depend on plant trait values in combination with the statistical properties of rainfall. The disadvantage of the stochastic model is that its assumptions sometimes cannot match reality. The capacitance-resistance

model uses Ohm's law to simulate the water transport process in plants; that is, it uses water flow, water potential, water transport resistance, and water storage function in plant tissues as the electric current, voltage, resistance, and capacitance, respectively, and establishes mathematical ordinary differential equations for scaling up the hydraulic traits of tissues to the hydraulic conductivity of the whole plant (Deng et al., 2017). Meanwhile, vulnerability curves, which describe xylem resistance to embolism, are often used to determine plant death. The capacitance-resistance framework has been widely used since it was proposed because it is close to the real water transport process. However, because it uses physical theory to explain life phenomena, it is insufficient to describe some nonlinear vegetation dynamics (Li, 2014). In terms of mathematical model methods, the tissue-to-plant upscaling of functional traits depends on a series of endogenous plant properties (e.g., allometry); the upscaling from individual to stand depends on the community-level distribution of these hydraulic traits, which can be affected by ecological (e.g., competition) and environmental (e.g., fertility, climate) factors; and the upscaling from stand to grid cell averages (in models) depends on the effects that hydraulic traits have on the long-term dynamics of stand development (recruitment, growth, competition, mortality) (Mencuccini et al., 2019).

The parameter equivalent method usually generalizes the small-scale process into a large-scale state by looking for the performance of small-scale behavior on a larger scale and replaces the small-scale process with a large-scale "equivalent parameter". Unlike the empirical statistical method, the parameter equivalent method links two scales through theoretical relationships rather than statistical relationships. For example, Anderson et al. (2003) believed that when there was no environmental stress, canopy-scale vegetation transpiration was mainly controlled by radiation rather than stomatal behavior. Therefore, when calculating canopy resistance, they used the analytical solutions of canopy-scale light use efficiency (LUE) and photosynthetic effective radiation to replace the numerical solution of the upscaling result through a leaf-scale photosynthetic equation; the results of the two methods agree well, but the method using LUE is more efficient and stable. In fact, some optimal theory models also belong to the parameter equivalent method. These models, in essence, express the upscaling result of tissue-scale stomatal behavior with the observed canopy-scale parameters using the "ecological optimization principle" or "microeconomic principle" (Bonan et al., 2014).

3.2 Spatial upscaling of soil hydrological process

The upscaling study of soil hydrological processes has always been a hotspot and difficult in the field of hydrology

and hydropedology because the soil porous media structure is uneven on all spatial scales (Pachepsky and Hill, 2017). At present, some progress has been made in multiscale equations for soil water flow and spatial upscaling for soil hydraulic parameters. For example, Roth (2007) proved that under the condition of hydrostatic equilibrium, the Navier Stokes equation describing pore-scale fluid movement is physically consistent with Darcy's law and the Richards equation describing profile-scale water movement. On a larger scale, most people believe that the Richards equation has the same mathematical form as on the profile scale, but its hydraulic conductivity needs to be redefined as an effective or equivalent parameter (Li, 2014). Therefore, the Richards equation is often applied on the regional scale in many LSM and hydrological models. Yang et al. (2014) developed a unified multiscale model (UMSM) that modifies the Navier Stokes equation by adding a Darcy force term to formulate a single set of equations, and therefore, it can be applied at multiple scales from the pore to ecosystem scales. Scheibe et al. (2015) developed a hybrid multiscale model that initially simulates the full domain at the continuum scale and applies a pore-scale model only to areas of high reactivity, increasing the accuracy of subsurface reactive transport simulations. Yang X F et al. (2021) reviewed current multiscale models of water flow and solute transport in porous media (such as soil) and discussed future research perspectives and challenges from both modeling and application points of view.

When soil water flow at different scales can be described by cross-scale mathematical equations, how to describe the spatial heterogeneity of soil hydraulic parameters and how to attain area hydraulic parameters through point sampling will be the key to the spatial upscaling of the soil hydrological process. At present, Miller's scaling theory, geo-statistic method, and fractal theory are the most widely used approaches for describing the spatial heterogeneity of soil characteristics, and the main upscaling approaches include stochastic perturbation methods, the aggregation concept, the stream-tube approach, inverse modeling approaches, and the data fusion method (Vereecken et al., 2007).

Scaling theory is based on the spatial similarity assumption of pore space structure (Miller and Miller, 1956, Whitaker, 1999). In this theory, two soils or porous media are similar when "scale factors" can be found that transform the behavior of one of the porous media to that of the other. It describes the spatial variation of soil characteristics by determining appropriate "scale factors" for different points in space and calculating the statistical characteristics and spatial correlation structure of these "scale factors" (Jia et al., 2004). However, its application is often limited to the field scale because it requires extensive measurement campaigns using classical sampling techniques to obtain detailed information on the variability of soil hydraulic properties

(Vereecken et al., 2007). The geostatistical method is based on the theory of regionalized variables and describes the hydraulic parameters as a function of location (Song et al., 2004). Within a geostatistical framework, a parameter, e.g., K_s , is considered a random realization of a random space function (RSF) $V(x)$. Typically, the spatial structure of $V(x)$ is defined either by the so-called semivariogram or by the correlogram. The geostatistical information is then used as part of an upscaling procedure (such as the kriging interpolation method or stochastic method) to predict effective hydraulic parameters, upscaled state variables, and fluxes (Wang et al., 2012). Compared with traditional statistical methods, geostatistical methods not only consider the irregularity (randomness) of the spatial distribution of soil hydraulic parameters but also consider the spatial correlation (determinacy) within a certain range (Wu S L, 2014). Using such (geo)statistical descriptors of a space variable as input, the so-called stochastic perturbation approaches allow us to statistically quantify upscaled properties, that is, quantify the mean state of the system at the larger scale and the local variance around the mean state at the larger scale (Vereecken et al., 2007). When the parameters of the local-scale hydraulic properties are treated as random space functions, the equations describing soil water flow are stochastic continuous equations; therefore, the moments (i.e., mean, variance, etc.) of the probability distribution of the state variables (such as soil water content, water potential, water flow) in the equation can be computed (Vereecken et al., 2007; Wu S L, 2014). However, all perturbation approaches are only accurate for mildly variable soil water conditions and can hardly be applied to highly heterogeneous soils.

At a larger scale, the aggregation concept allows us to describe soil water processes in soil map units or numerical grids in LSM and climate models (Vereecken et al., 2007). It is assumed that the soil inside an aggregation unit can be equivalently replaced by homogeneous soil. Under unsaturated conditions, soil water flow is mainly vertical, so the water flow of each “aggregation unit” has the characteristic of additivity (Vogel, 2019). The soil physical and chemical properties of each aggregation unit can be obtained by the “average composition method” or “dominant fraction method”, and then the soil properties (such as texture, bulk density and soil organic matter) are converted into soil hydraulic parameters through Pedotransfer functions (PTFs) (Dai et al., 2013). Finally, the vertical flow of each aggregation unit is added to obtain the larger-scale soil flow process. A significant shortcoming of the aggregation concept approach is its unreliable prediction of ET, percolation, and root water uptake. In fact, this approach amounts to a first-order approximation of stochastic models. It therefore ignores the effective contributions to soil water fluxes from parameter and process interactions (in space and time) in a heterogeneous system (Vereecken et al., 2007). Other up-

scaling approaches will not be discussed here due to word count limitations.

Overall, the upscaling studies mentioned above mainly focus on unsaturated vertical soil water flow and depend on certain assumptions. However, the real-world soil hydrological process is affected by many other factors, such as topography and root dynamics; there are practically no datasets presently available that provide sufficient information to extensively validate existing upscaling approaches. Hydrogeophysical methods combined with remote sensing techniques promise significant contributions toward providing high-quality datasets (Vereecken et al., 2007).

3.3 The relationship of the time scale to the spatial scale in the study of coupled soil-vegetation-hydrology

This article reviewed coupled processes and mechanisms of soil-vegetation-hydrology at different spatial scales, but the time scale related to these processes is not explained. Here we briefly discuss the relationship of the time scale to the spatial scale.

Essentially, processes acting in short-term can be summarized over longer timespans (for example, daily surface flow can be summed over a year, and then be used for annual erosion calculations), or multiplied with the scaling factor to bridge the different timescales of the processes (van der Meij et al., 2018). Therefore, a process that occurred in a certain spatial scale can be studied from multiple time scales in theory. However, current studies of coupled soil-vegetation-hydrology are usually limited by observation means, which usually have a fixed spatial and temporal scale. For example, in the pedon to hillslope scale, the observation system for soil, vegetation and hydrological processes are often on a minute to hour time scale—such as eddy covariance towers, automatic meteorological stations, and sensor network of soil moisture and leaf area index. In the catchment scale, the study of coupled soil-vegetation-hydrology often relies on satellite-based data, which normally have a time scale from one week to one month.

In general, for most of the coupling processes of soil-vegetation-hydrology, the larger the spatial scale, the longer the time scale (Figure 2). Meanwhile, the time scale of research on soil processes is usually much longer than that of ecological and hydrological processes. For example, the processes of water and carbon exchange in stomatal scale can varied in minutes to hours scale (Fatichi et al., 2016); in the whole plant scale, the water balance, plant physiology, and soil properties are usually investigated and studied in the time scales of hours, days, and decades respectively. When it comes to the basin scale, the study of daily-scale hydrological processes and yearly-scale vegetation dynamics (such as recruitment, growth, competition, and mortality) is more common, and in these cases, soil characteristic is often

treated as constant parameters (Mencuccini et al., 2019).

4. Modeling approaches for coupled soil-vegetation-hydrological processes

Models, as an abstract and simplification of real physical processes, are an important means to study the coupling mechanism of soil-vegetation-hydrological processes. Sometimes, modeling approaches are even the only research means available in cases that lack observation data. The development of models for coupled soil-vegetation-hydrological processes can be divided into three stages: (1) in early studies, scholars in different fields mostly studied the coupled soil-vegetation-hydrological process separately and developed models of individual processes that were only related to their own discipline. (2) With the deepening of research and the development of disciplines, scholars gradually began to develop multiscale and multiprocess integration models. (3) Different integration models have different emphases on module composition, and sometimes it is difficult to meet special actual needs, and therefore, some scholars have coupled different integration models to develop a multimode coupling model.

4.1 Models of individual process

Since the establishment of modern disciplinary systems in the 18th century, researchers from different disciplines have

focused on their own fields and have explored coupled soil-vegetation-hydrological processes only from the perspective of a single discipline. As mentioned above, at different spatial scales, the dominant coupling mechanisms of soil-vegetation-hydrological processes are totally different. For the six coupling processes summarized in this study, most of the early developed models only take one or several processes as the object and simulate them in detail. Such models often investigate physical processes heavily based on abundant experiments and observations. With the exploration of individual coupling processes, the old models are constantly optimized, and new models are constantly proposed (Table 2).

Taking the infiltration model as an example, modeling research for the soil infiltration process has started and has been continuously improved since the middle of the 19th century. In 1856, Henri Darcy, a French engineer, initially developed Darcy's law describing saturated infiltration through experiments for municipal construction; Green and Ampt (1911) established the Green-Ampt infiltration model with a certain physical mechanism in 1911; Horton (1941) and Philip (1957) established two classical infiltration models in 1941 and 1957, respectively; in the following half a century, a series of newly developed infiltration models (such as Zhang's model for analyzing rainfall infiltration in slopes (Zhang J et al., 2016)) were proposed for different environmental conditions. Modeling research on stomatal behavior has also experienced a similar history—from the Jarvis empirical model (Jarvis, 1976) and Ball semiempirical

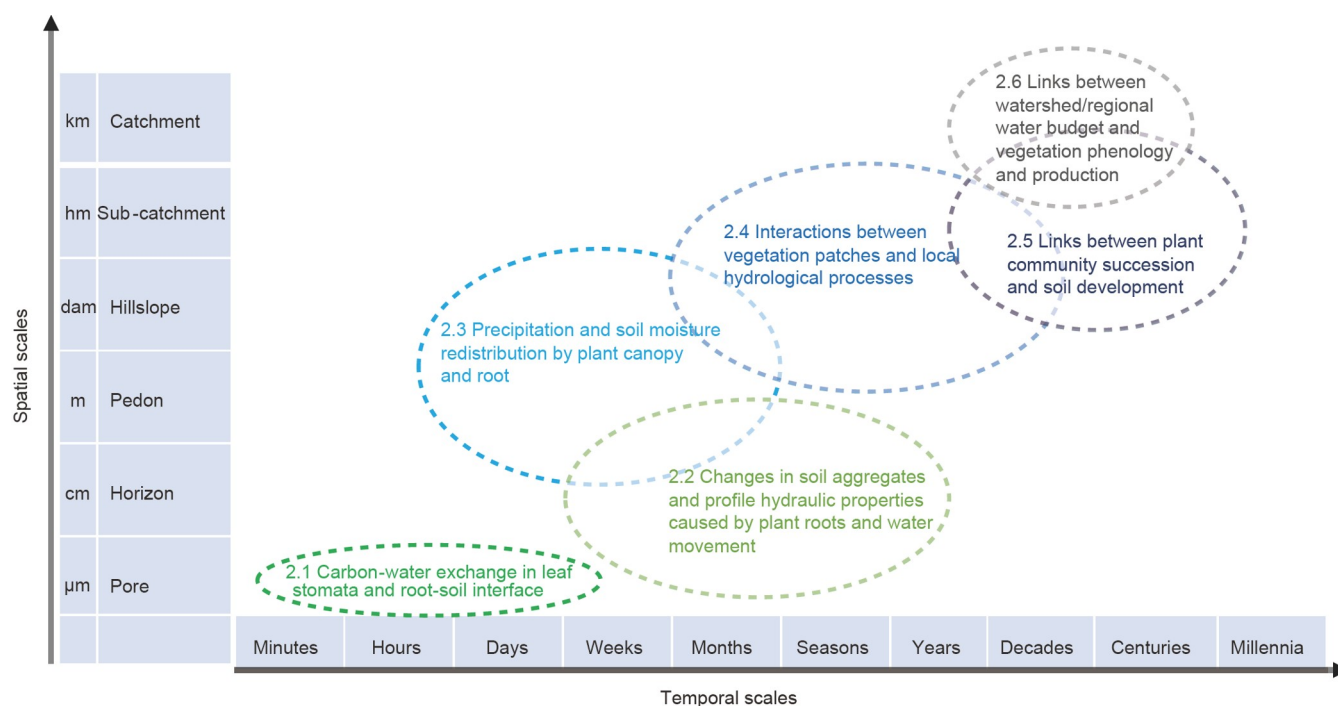


Figure 2 A sketch of the spatial and temporal scales over which the effects of different types of processes, represented by different ellipses, become apparent.

Table 2 Classical models of individual process^{a)}

Processes	Models describing the corresponding process
①	Water transport in xylem: Sperry et al. (1998) ; TREES model (Loranty et al., 2010); Stomata behavior: Jarvis (1976) ; Ball et al. (1987) ; Gao et al. (2002) ; Nutrient transfer in the soil-root system: Silberbush and Barber (1983) ; Darrah (1993) ; Zhang et al. (2002) ; Rhizosphere microorganism-root interaction: Marschner et al. (2011) ; Rhizosphere soil-root physical interaction: Wang X et al. (2020)
②	Soil aggregates-roots feedback: Misra et al. (1988) ; Soil aggregates formation: Oades (1984) ; Plante et al. (2002) ; Liu K et al. (2020) ; Clay translocation: Vanwalleghe et al. (2013) ; Campforts et al. (2016) ; LORICA model (Temme and Vanwalleghe, 2016); Soil hydraulic properties: Soil water retention curve: Brooks and Corey (1964) ; van Genuchten (1980) ; Hydraulic conductivity: Burdine (1953) ; Mualem (1976) ; Pedotransfer functions (PTFs): Cosby et al. (1984) ; Saxton et al. (1986) ; Saxton and Rawls (2006) ; Soil hydrological process: Buckingham-Darcy law (Buckingham, 1907); Richards equation (Richards, 1931); MODHMS model (https://www.hgl.com/softwareproducts-new/modhms , 2000); HYDRUS 2D/3D model (https://www.pc-progress.com/en/default.aspx?hydrus-3d , 2012)
③	Canopy interception: Rutter et al. (1971) ; Gash (1979) ; Throughfall and stem flow: Wang et al. (2005) ; Infiltration process: Green and Ampt (1911) ; Philip (1957) ; Zhang J et al. (2016) ; Root water uptake: Vrugt et al. (2001) ; Schneider et al. (2010) ; Warren et al. (2015) ; Root hydraulic lift: Landsberg and Fowkes (1978) ; Mendel et al. (2002) ; Yu and D'Odorico (2014) ; Plant response to drought stress (Crop growth model): CERES model (https://dssat.net/models-overview/components/ceres-wheat , 1984); WOFOST model (Van Diepen et al., 1989); AquaCrop model (Steduto et al., 2009)
④	Formation of vegetation patch pattern: Klausmeier (1999) ; von Hardenberg et al. (2001) ; Liu et al. (2008) ; Rietkerk and van de Koppel (2008) ; Feedback between vegetation patch and hydrological process: Entekhabi et al. (1992) ; Dekker et al. (2007) ; Hwang et al. (2009) ; Soil water erosion: USLE model (Wischmeier and Smith, 1965); WEPP model (Ascough et al., 1997); RILLGROW model (Favis-Mortlock, 1998); DYRIM model (Wang et al., 2007)
⑤	Plant-soil feedback: Herbert et al. (1999) ; Miki and Kondoh (2002) ; Marasco et al. (2020) ; Soil evolution/soil profile formation: Kirkby (1977) ; SoilGen2 model (Finke, 2012); COMMISSION model (Ahrens et al., 2015); Kirkby (2018) ; Vegetation community succession: FOROAK model (Shugart and West, 1977); NEWCOP model (Zhao et al., 1998); GFSM model (Chen and Luo, 2002)
⑥	Watershed hydrological process: TOPMODEL model (https://geomodeling.njnu.edu.cn/modelItem/5ee7cd6a-c71d-45a8-b7d6-060c07978be9 , 1979); SHE model (Abbott et al., 1986); SWAT model (Arnold et al., 1998); Ecosystem carbon/water/energy cycle: Thornthwaite Memorial model (Lieth and Box, 1972); PnET model (Aber and Federer, 1992); BIOME-BGC model (Running and Hunt, 1993)

a) Processes ①–⑥ are: ① Carbon-water exchange in leaf stomata and root-soil interface; ② Changes of soil aggregates and profile hydraulic properties caused by plant roots and water movement; ③ Precipitation and soil moisture redistribution by plant canopy and root; ④ interactions between vegetation patches and local hydrological process; ⑤ links between vegetation community succession and soil development; ⑥ links between watershed water budget and regional vegetation production

model ([Ball et al., 1987](#)) at the beginning to the mechanism models considering ABA and guard cell pressure (e.g., [Tardieu and Davies \(1993\)](#), [Dewar \(1995\)](#)) as well as the optimal stomatal conductance model (e.g., [Medlyn et al. \(2011\)](#)).

Meanwhile, there are still some individual processes that lack relevant models. For instance, research on the feedback process between soil aggregates and plant roots is largely experimental research, and there is a lack of effective modeling approaches thus far.

4.2 Multi-scale and multi-process integration models

With the increase in spatial scale, the interaction among soil, vegetation, and hydrology becomes increasingly intense and complex, and a change in one of them can significantly influence the other two. As a result, multiscale and multi-process integration models have been gradually developed to accurately simulate coupled soil-vegetation-hydrological processes at a large scale. The most common multiprocess integration models ([Table 3](#)) include (1) the comprehensive ecohydrological model developed based on the hydrological process model in the hydrology field; (2) the terrestrial biosphere model (TBM) (ecosystem model) developed from the plant community dynamic model in the ecology field; (3) the land surface model (LSM) developed for improving the simulation accuracy of land-atmosphere exchange; and (4)

the Earth system model (ESM) developed from a climate system model that involves contributions from many scientists and many areas of science ([Flato, 2011](#)).

These multiprocess integration models usually integrate the classical individual process model at different scales into one unified framework to construct a comprehensive mechanism that can accurately describe large-scale coupled soil-vegetation-hydrological processes. In this process, model developers tend to choose these simple, reliable, and widely used individual process models, which were usually proposed years ago and cannot represent the latest research results in their fields. Therefore, in terms of simulation accuracy and physical mechanism, the latest individual process models are usually ahead of the multiprocess integration models. For example, in most multiprocess integration models, stomatal behavior is determined by only a few simple environmental factors, and its direct feedback on the vascular hydraulic system is rarely considered ([Mencuccini et al., 2019](#)). In CLM4.5 (LSM model), stomatal conductance was calculated by the Ball-Berry semiempirical model ([Ball et al., 1987](#)), in which stomatal conductance is controlled only by photosynthetic rate, relative humidity, and CO₂ concentration. In contrast, in the study of tissue-scale plant physiology, stomatal behavior can be directly impacted by many other factors, such as soil water content, air temperature, light intensity, xylem water potential, abscisic acid concentration, and other plant chemical or biological factors

Table 3 The most common multi-scale and multi-process integration models

Model type	Examples
(1) Watershed Ecohydrological model	EcoHAT; TOPOG; RHESSys; VIP; Macaque; BEPS-TerrainLab
(2) Terrestrial Biosphere model (TBM)	CoupModel; LPJ-GUESS; ORCHIDEE; CABLE; DLEM; JULES
(3) Land Surface model (LSM)	NOAA; CoLM; CLM; UKMO JULES; BATS; SIB; Noah-MP
(4) Earth System model (ESM)	CanESM2 (Canada); MPI-ESM (Germany); MIROC-ESM (Japan); GFDL-ESM2M (U.S.A), EC-EARTH (Europe)

(Jiang et al., 2020). Bonan et al. (2014) evaluated the Ball-Berry model used in the CLM4.5 and an alternative optimal stomatal conductance model considering plant hydraulic constraints and found that the optimal model was significantly better than the CLM Ball-Berry model. The LPJ-GUESS model still uses the theory proposed by Sykes and Cramer (1996) to simulate vegetation phenology (considering only chilling and GDD demand), which is not completely consistent with many multi-process and multi-factor driven phenology models proposed in recent studies (Liu et al., 2018). Another persuasive example is the root water uptake process (and the corresponding nutrient uptake). In ecohydrological models and Earth system models, the most common methods to calculate root water uptake (or transpiration) are to link it with soil water content (or water potential) through an empirical relationship under a given root depth (zero-dimensional model) (Warren et al., 2015; Fatichi et al., 2016); other methods consider the vertical distribution of the root system (one-dimensional) (Fatichi et al., 2012), and build rough soil-root conductivity by using the root length or root area index (Daly et al., 2004); however, nearly no three-dimensional root module exists in ecohydrological models and ESM, although some complex three-dimensional root water uptake models, which can qualitatively and quantitatively describe root-soil water transport, have been developed in the plant hydraulics field (Vrugt et al., 2001; Doussan et al., 2006; Javaux et al., 2008).

At the same time, the existing multiprocess integration models are basically developed from one major subject (field) and therefore ignore some of the coupled soil-vegetation-hydrological processes that are emphasized in other subjects (fields) (Table 4). For example, in most terrestrial biosphere models, simulations of soil physical and microbial processes lack soil aggregate formation processes and can hardly accurately simulate the impact of microbial processes on ecosystem C, N, and P cycles (Peng et al., 2020). The existing LSM generally use empirical parameters to describe plant hydraulics, which make it difficult to simulate the response of plant physiology to environmental stress (Eller et al., 2020); meanwhile, most LSM do not consider two-dimensional or three-dimensional hydrological processes (Fatichi et al., 2016), that is, they are not able to simulate lateral hydrological processes and thus the real watershed runoff process. The existing ecohydrological models and

ESM also lack the mechanism of “vegetation-hydrology feedback” from the perspective of plant physiology, resulting in an incomplete and inaccurate description of the dynamics of coupled soil-vegetation-hydrological processes (Chen et al., 2011; Fan et al., 2016).

The accuracy of large-scale soil-vegetation-hydrology modeling is affected by many factors, including the model structure, the model parameter errors (e.g., ESM may have hundreds of parameters), and the forcing data quality. In order to improve the accuracy of estimation, researchers assimilate the remote sensing data and in-situ observation data into models, developing the land data assimilation system (LDAS). Data assimilation techniques include Kalman filtering, Bayesian filtering, Optimal interpolation, Genetic algorithm, Variational method, and so on. The evaluation metrics for validate and evaluate the LDAS products include bias, relative bias, root-mean-square-error (RMSE), and Nash-Sutcliffe efficiency, based on different spatio-temporal scales (hourly to yearly, sites to continent) (Xia et al., 2019). Since the establishment of North American LDAS in 2004, Data Assimilation methods have been widely used in meteorological models, climate models, drought monitoring, and crop and water resources management.

4.3 Multi-mode coupling model

As mentioned above, multiprocess integration models are usually developed from one major subject (field); different integration models have their own advantages as well as drawbacks. In some special application scenarios, such as weather numerical simulation, climate prediction, and watershed water management, multiprocess integration models may not meet the demand for accuracy. Therefore, some scholars have coupled different integration models together and developed multimode coupling models. These models account for missing processes and gaps in individual integration models to some degree and are more applicable for some special studies.

For example, Etchevers et al. (2001) coupled the ISBA model (LSM) with the MODCOU model (hydrological model), resulting in a better estimation of the daily-scale hydrological cycle and energy budget. Maxwell and Miller (2005) coupled CLM (LSM) with ParFlow (distributed hydrological model) and found that the simulated soil moisture

Table 4 Module comparison of different multi-process integration models^{a)}

Coupling process	CLM5.0	Noah-MP	CoupModel	ORCHIDEE	LPJ-GUESS	EcoHAT	RHESSys
①	√	○	√	√	√	○	○
②	○	×	○	×	○	×	×
③	○	○	√	○	√	√	√
④	○	○	○	○	√	○	√
⑤	○	○	○	○	○	○	○
⑥	√	√	√	√	√	√	√

a) This table aim to show the advantages and disadvantages of different multi-process integration models in simulating the coupling of soil-vegetation-hydrological processes (CSVHP), where CLM5.0 and Noah-MP belong to LSM, CoupModel, ORCHIDEE and LPJ-GUESS are TBMs, and the other two are watershed Ecohydrological model. The Coupling Process ①–⑥ are the same as Table 2. The symbol “√” indicates that the model contains modules that can simulate most of the mechanisms in the corresponding CSVHP, the “○” indicates that the model contains modules that can simulate limited mechanism in the corresponding CSVHP, and the “×” indicates that the model cannot simulate the corresponding CSVHP.

and groundwater level were in good agreement with the observed values. Niu et al. (2007) coupled CLM with SIMGM (groundwater model) to simulate the changes in global surface runoff and water storage amount. Niu et al. (2011) coupled Noah (LSM) with the SIMGM model, improving the simulation accuracy of the Noah model for soil water and evapotranspiration. Lam et al. (2011) coupled MODFLOW (hydrological model) with LSM to calculate the water budget in the study area, showing that the contribution of groundwater to evapotranspiration is significant. Zampieri et al. (2012) coupled CLM 3.5 with TOPMODEL (hydrological model) and found that rivers have a significant impact on regional soil moisture and surface temperature patterns. The study of Maxwell and Miller (2005) showed that LSM coupled with three-dimensional hydrological processes have more advantages than uncoupled models in simulating soil moisture, latent and sensible heat, temperature, and runoff and in predicting groundwater depth.

Cai et al. (2016) coupled FUN's (plant model) plant nitrogen uptake module and SWAT's (water management model) soil nitrogen cycling module into Noah-MP; the results showed that the coupled model, i.e., Noah-MP-CN, performed well in capturing the major nitrogen state/flux and advanced the capability of Noah-MP to simultaneously predict weather and water quality. Yang et al. (2019) introduced the nitrogen cycle process into the AVIM (Atmosphere-Vegetation Interaction model) and developed a new AVIM-CN model that contains the carbon-nitrogen coupling process, significantly improving the simulation accuracy of the carbon flux of terrestrial ecosystems. In view of the insufficient description of the vegetation-hydrology feedback mechanism in existing multiprocess integration models, Dekker et al. (2007) added the large-scale evapotranspiration-precipitation feedback mechanism and the small-scale vegetation-infiltration feedback mechanism into a SPAC model; it was found that the elasticity (resilience) of ecosystems to climate change is changed significantly after adding different scale feedbacks.

At the landscape scale, Gao et al. (2018) coupled a plant

growth model (ALMANAC, which takes plant competition into account) with SWAT model to improve the accuracy of simulations of plant growth on hydrological and biogeochemical cycles. Van der Meij et al. (2018) added the hydrological process considering terrain into a soil-landscape evolution model (Lorica), and found that, in a modified Lorica model, the increased clay translocation in the surface depressions, caused by higher infiltration rates, can better match the field evidence in pedogenesis. Van Oost et al. (2005) incorporated a soil carbon dynamics model (ICBM, including both water- and tillage-induced soil redistribution) in a soil-landscape evolution model (SPEROS), which improved the prediction ability of soil organic carbon pattern.

It should be noted that the way of coupling is different in different studies; most of them are tight coupling (bidirectional coupling) while the others are loose coupling (one-way coupling), for example, in the study of Coupled model CLM.PF, the CLM, and ParFlow models were coupled at the land surface and soil column by replacing the soil column/root-zone soil moisture formulation in CLM with the ParFlow formulation. However, in the case of coupling ISBA and MODCOU, surface runoff and gravitational drainage are first computed by ISBA and then transferred to MODCOU; there is no feedback from MODCOU to ISBA (Etchevers et al., 2001). In the case of coupling CLM with SIMGM, “coupling” means adding an underlying unconfined aquifer module, i.e., SIMGM, to CLM's soil column, and the interaction between SIMGM and CLM is accounted (Niu et al., 2007).

4.4 New trends and challenges of models of coupled soil-vegetation-hydrological processes

4.4.1 Other type of models

Traditional models of coupled soil-vegetation-hydrological processes are generally based on either empirical relationship or physical mechanisms. Both of them have limitations: the application of physical mechanism-based models may be limited by parameter acquisition and computing power,

while the empirical models are usually not applicable for broader regions. Therefore, in addition to these traditional models, some scholars have begun to use other type of models like a stochastic model, neural network model, and optimal theory model to simulate the coupling process between soil, vegetation, and hydrological processes.

Eagleson (1978) first introduced the stochastic model into ecohydrological process modeling: he established a soil hydrodynamics model driven by random meteorological conditions, and bridged the soil wetness and optimum vegetation canopy density by differential equations deduced from equilibrium hypotheses, finally explored a statistical-dynamic model with respect to the hydrologic parameters of both soil and vegetation (Eagleson and Peter, 1982). Since then, Milly (1993), Rodriguez-Iturbe et al. (1999), Porporato et al. (2002), and Laio et al. (2006) have continuously improved and developed their own stochastic ecohydrological models, and connected the infiltration process, soil moisture dynamics and plant water stress with carbon assimilation. These kinds of stochastic models can be used to evaluate the impacts of climate, soil, and vegetation on soil water probability density function and long-term average water balance. In addition, in the ecology field, some scholars had used stochastic models to simulate vegetation species competition and population dynamics. For example, Duan (2015) used stochastic differential equations to describe population dynamics. Compared with the physical mechanism-based model, the stochastic models provide a more efficient way to describe the ecohydrological process with fewer variables and parameters, resulting in a more concise model structure.

At present, artificial neural network (ANN) method has been widely used in the hydrological simulation of runoff, the prediction of precipitation, the construction of Pedo-transfer Functions, and the estimation of soil moisture, evapotranspiration and vegetation index using remote sensing products (Liu, 2011; Hassan-Esfahani et al., 2015; Van der Meij et al., 2018). For example, in some cases, the accuracy of soil moisture estimated by ANN model is even higher than that estimated by physical process-based model (Elshorbagy and Parasuraman, 2008). One of the drawbacks of ANN methods is that it is very dependent on the selection of input variables. The optimal theory-based models, which have been mentioned above, use a few parameters to determine the key ecohydrological state and, therefore, can significantly simplify the model framework. However, the optimal behavior stems from the perspective of a long-term equilibrium state and may not be valid when the environment changed suddenly.

4.4.2 Limitations and research frontier of models of coupled soil-vegetation-hydrological processes

The limitations of the existing models for coupled soil-ve-

getation-hydrological processes include but are not limited to: (1) the existing models mainly focus vegetation-hydrological processes interactions yet without considering the soil physical processes. Previous studies showed that accounting for soil physical process mechanisms in TBMs can significantly improve the simulation accuracy of carbon, nitrogen, and water cycles (Peng et al., 2020). However, TBMs, LSM, and Ecohydrological models are more or less composed by modules simulating vegetation dynamics, hydrological processes and their interactions (Fatichi et al., 2016), lacking of modules that can characterize soil physical processes—such as the soil aggregate formation process and the temporal dynamics of profile hydraulic properties. Similarly, in the soil-landscape evolution model, it is difficult to couple the hydrological and plant process at the daily timescale with the soil evolution process at the millennial timescale (Van der Meij et al., 2018). (2) In nature, the spatial and temporal pattern of vegetation community and structure in ecosystem is dynamic, especially in the context of climate change, and the stand structure of terrestrial ecosystem is closely related to the environment and constantly changes dynamically (Gonzalez et al., 2010; Higgins and Scheiter, 2012), while most of the current models, even TBMs, do not take this into account. As a link between soil and atmosphere, accurate simulation of interannual fluctuation of ecosystem stand structure is of great significance for a deep understanding of the coupled soil-vegetation-hydrological process. (3) At a large scale, vegetation dynamics and hydrological processes are closely related to atmospheric circulation, and vice versa. However, most existing models only employ atmospheric conditions as force, and do not consider the feedback effect of vegetation change on the atmospheric water cycle (Peng et al., 2005). The bidirectional feedback mechanisms of vegetation-hydrology or atmosphere-hydrology remain to be explored (Ning et al., 2019), which limits our capability to predict the effect of climate change on vegetation distribution and its feedback on the global water cycle and climate (Eller et al., 2020). (4) Lacking of generalized parameterization schemes that can well describe the impact of human activities on soil, vegetation, and hydrological process, which would lead incomplete or biased simulation results (Liu G et al., 2020). In addition, because of the difference in time scales between vegetation (or hydrological) processes and soil processes, as well as the associated uncertainties of parameters and the ignorance of sub-grid physical process (Ning et al., 2019), large-scale and high-resolution simulation for coupled soil-vegetation-hydrological processes is still a challenge.

The research frontier of models of coupled soil-vegetation-hydrological processes includes: (1) the error estimation and automatic optimization of model parameters. Due to soil spatial heterogeneity, vegetation temporal dynamics, limited monitoring stations and other reasons, the determi-

nation and calibration of soil, vegetation, and hydrological parameters is not easy when using multi-process integration models (Chen et al., 2011), which usually contains a large number of parameters, and may cause the inaccuracy of simulation result (Ning et al., 2019; Gao and Zhao, 2020; Sun et al., 2022). Using optimization algorithms to automatically optimize and estimate parameters in models has gradually become a hotspot (Zhang T, 2016). (2) Developing a more comprehensive model of coupled soil-vegetation-hydrological processes. The coupling mechanism of soil-vegetation-hydrology that exist in LSM, TBMs, and Ecohydrological models is becoming more and more similar, and their differences may gradually disappear in the future (Fatichi et al., 2016). Therefore, it should be possible to develop a more rational and comprehensive multi-process integration model based on the state-of-the-art research in the fields of soil, vegetation, and hydrology, and improve the simulation accuracy through methods like data assimilation (Li et al., 2020) in the future. Figure 3 provided a conceptual design to couple some, but not all, existing knowledge about soil-vegetation-water feedback process into a multi-scale and multi-process integration model. At the same time, we have to admit that developing a new integration model that is comprehensive enough to include most of the coupling processes of soil-vegetation-hydrology is quite difficult, and it requires long-term collaboration and researchers from multiple disciplines at different regions and countries. (3) Developing data-driven model and module-selectable model. With more and more available data sets of soil, vegetation and hydrological processes, data-driven model with few parameters has become one of the most potential approaches to quantify ecological and hydrological processes (Breña Naranjo et al., 2011). Module-selectable model is also popular because of its flexibility, for example, users can choose either Priestley-taylor model or Penman-Monteith model to calculate Evapotranspiration, and choose either GLOBFIRE model (Knorr et al., 2014) or GLOBFIRE-BLAZE model (Hamilton et al., 2018) to simulate wildfire, according their needs.

5. Current limitations and future work

5.1 Current limitations

5.1.1 Limitations on the observations of coupled soil-vegetation-hydrological processes

Advances in observation means or monitoring technologies can significantly improve the understanding of coupled soil-vegetation-hydrological processes, and the representative observation limitations include the following: (1) it is difficult to continuously monitor the key chemical substances affecting intrinsic plant water transport, such as abscisic acid (ABA) and aquaporins (Wang Z, 2021). (2) There is a lack of

noninvasive, real-time, and *in situ* investigations of local rhizosphere processes, which limits the integration of the multiple root-soil interaction processes occurring in the rhizosphere (Wang X et al., 2020). (3) Monitoring three-dimensional root dynamics in field conditions is still an immense challenge, which limits the studies of root functioning and the development of models or conceptualization. (4) There is a lack of direct observations of rock moisture, i.e., the exchangeable water stored in the unsaturated zone in weathered bedrock (Rempe and Dietrich, 2018). (5) Observation data of soil hydraulic properties in high-altitude mountainous areas are sparse, and the vertical distribution information of soil hydraulic properties in many areas is insufficient (He et al., 2021). (6) Building a multiscale monitoring network of hydrological processes and soil hydraulic properties is limited by cost (He and James, 2021).

5.1.2 Knowledge gaps in the mechanisms of coupled soil-vegetation-hydrological processes

The representative knowledge gaps include the following: (1) for an individual plant, its physiological mechanism of plant death, hydraulic mechanism of xylem embolism (cavitation) repair, and chemical mechanism of stomatal behavior and root water uptake are not fully understood (Fatichi et al., 2016; Jin et al., 2016; Xu et al., 2019). (2) For studies at the profile or hillslope scale, little is known about the mechanisms governing root-rock moisture interactions, the role of terrain in hillslope soil-vegetation interactions, the temporal dynamics of canopy structure and soil hydraulic parameters, and the important but neglected soil micro-organisms involved in plant nutrient utilization (Ehrenfeld et al., 2005; Gutiérrez-Jurado et al., 2006; Wang X et al., 2020; Wang G X et al., 2021). (3) For studies at long time scales, equations that can mechanistically describe soil chemical or physical weathering processes are still immature, and the impact of climate change on the key coupling mechanism of soil-vegetation-hydrological processes remains uncertain because of the difficulty of accurately predicting future hydrological processes on a long time scale (e.g., 50 years) (Egli and Fitze, 2001; Li and Fang, 2016; van der Meij et al., 2018). In addition, some new coupling mechanisms of soil-vegetation-hydrological processes are under rapid development; for example, the latest research shows that water in the atmosphere can also be absorbed by leaves and enter the soil through the plant body to buffer the negative effects of soil moisture stress (Boanares et al., 2020).

5.1.3 Limitations on scaling methods of coupled soil-vegetation-hydrological processes

Scaling issues have attracted widespread interest in the fields of ecology, hydrology, and soil science; however, there is still a long way to go before obvious advances are made. The development of upscaling and downscaling methods is

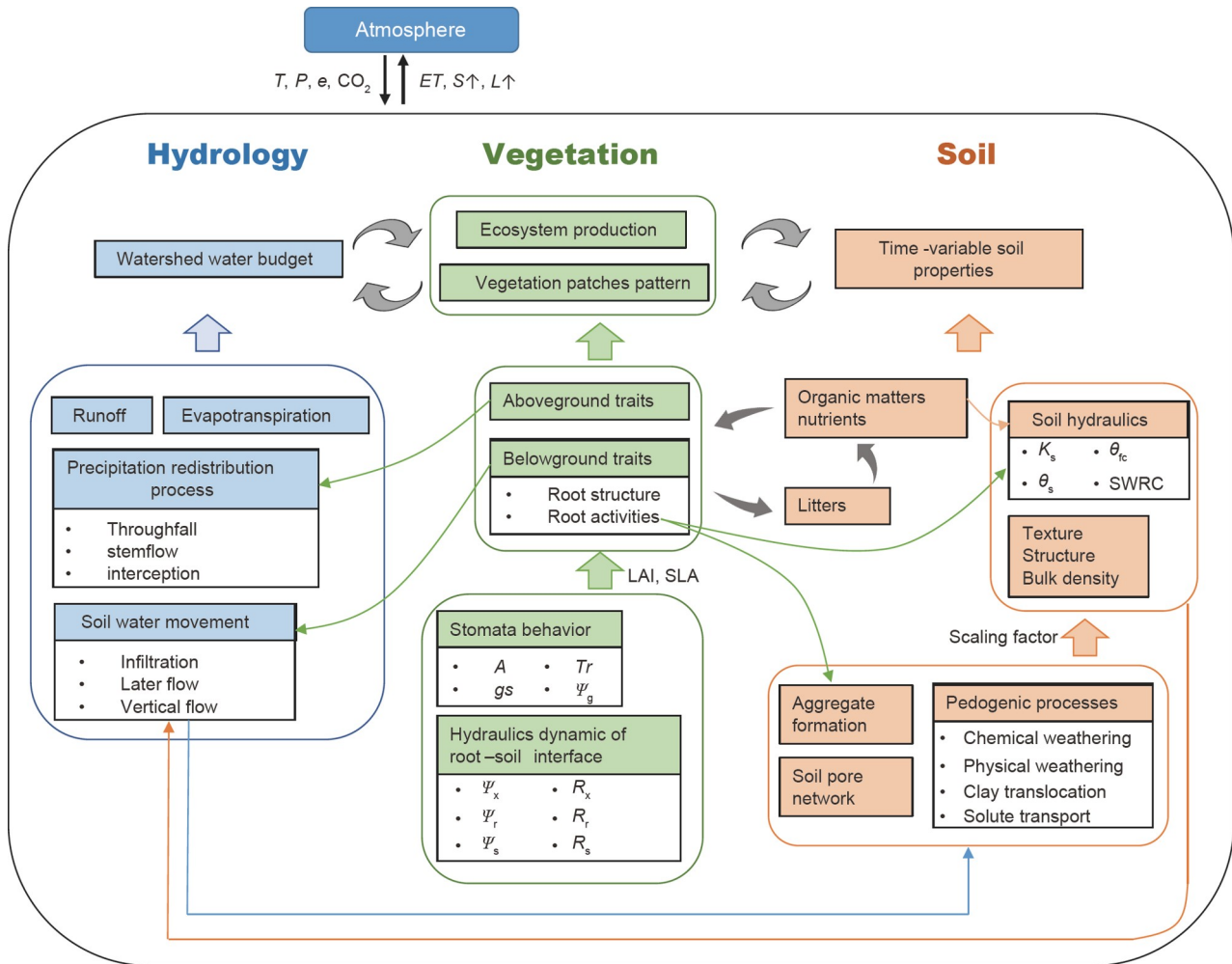


Figure 3 Conceptual design of a simple model of coupled soil-vegetation-hydrological processes, which integrate some existing knowledge about soil-vegetation-hydrology feedback process. The solid lines indicate connections (or feedbacks) between soil, vegetation, and hydrological process. The terms ψ_x , R_x , ψ_r , R_r , ψ_s , R_s is water potential and hydraulic conductivity of xylem, root, and rhizosphere soil respectively. The terms A , Tr , gs , ψ_g is photosynthetic rate, transpiration rate, stomatal conductance, and water potential of guard cell respectively. The terms T , P , e , ET , $S\uparrow$, $L\uparrow$ is air temperature, precipitation, saturated vapor pressure, evapotranspiration, upward shortwave, and upward longwave respectively.

limited by available information or datasets that can be used to validate existing scaling approaches. For example, scaling-up procedures from tissues to whole plants can be addressed mathematically, but the limited information on vertical variation in hydraulic traits, whole-plant architecture and allometry currently limits advances in this direction (Mencuccini et al., 2019). The stochastic method for soil water process upscaling usually assumes that hydraulic conductivities vary randomly, whereas we know that preferential flow paths often determine and control the response. Even though they do work, these upscaling methods are essentially ‘unverifiable’ at the catchment scale (Blöschl, 2001). The other limitation of the development of scaling methods is the difference in research scale in different disciplines and the shortage of mathematical algorithms for scale conversion. For example, changes in soil properties become apparent after 100 s or 1000 s of years, while soil hydrologic processes react subdaily to seasonal changes.

Somehow, these two processes are coevolving. However, these differences in process timescales complicate the coupling of soil and hydrological processes because hydrological processes are practically impossible to simulate over millennial time scales on a catchment scale (van der Meij et al., 2018).

5.1.4 Limitations on modeling approaches for coupled soil-vegetation-hydrological processes

In terms of model structure, assumption, and module composition, the limitations of existing models include the following: (1) only a few large-scale models consider the effect of root growth on soil hydraulic properties in a short period (Lu et al., 2020). (2) There is a lack of deep integration of different feedback mechanisms among soil, vegetation, and hydrological processes. (3) Some simple assumptions, such as optimal theory, are inconsistent with observations when the environment changes (Buckley and Schymanski, 2013).

(4) Physic-based theory may not well explain life phenomena, such as the plant dynamics in canopy interception and vegetation phenology simulation (Li, 2014). (5) The same or similar approaches have been used on very different scales at many LSM or TBMs; hence, there is a problem related to the ‘fallacy of average’, i.e., the false assumption that the mean of a nonlinear function of several variables equals the function of the means of these variables (Fatichi et al., 2016).

5.2 Future work

Studying the coupling of soil, vegetation, and hydrological processes is vital for understanding surface Earth systems and achieving sustainable development, attracting widespread interest among ecologists, soil scientists, hydrologists, and climate modelers. However, the observations, mechanisms, scaling methods, and modeling approaches of coupled soil-vegetation-hydrological processes still have many limitations. Future works should focus on:

(1) Achieving a deep integration of the coupled soil-vegetation-hydrological processes that exist at different spatiotemporal scales. In particular, a framework that can integrate “rapid ecohydrological process” and “slow soil process” is developed into one mathematical model is needed. To accomplish this objective, future work should strengthen multiscale, multifactor and multiprocess soil-vegetation-hydrology coupling observations and mechanism studies, the development of scaling methods, and the co-operation among different disciplines. Given that building and maintaining a multiscale monitoring network of soil-vegetation-hydrology need a mass of financial investments and hard labor, such challenging program may be best accomplished through a platform (such as the Critical Zone Observatory supported by the National Science Foundation of the United States), where the multiple scientific communities can cooperate to study the various processes at different spatial scales.

(2) Improving knowledge of short-term plant trait dynamics, which is a promising way to improve the understanding and modeling accuracy of coupled soil-vegetation-hydrological processes. Plant short-term growth dynamics can affect soil profile hydraulic properties (by root growth), canopy interception amount (by leaf and stem area change), and the applicability of the soil spatial scaling method (by root spatial distribution pattern). Future work should take this effect into account and explore appropriate ways to couple short-term plant trait dynamics into existing models—such as using a random variable.

(3) Paying more attention to the change in the coupling and feedback mechanism of soil-vegetation-hydrological processes under climate change and human activities and the eco-environmental effects brought by this change. If we do not consider the vegetation physiology variation caused by

climate change, it is impossible to make a persuasive long-term prediction of future ecosystem dynamics and the regional water cycle.

(4) Identifying different feedback pathways and integrating them into existing models. Numerous studies have shown that adding feedback mechanisms of “soil-vegetation” or “vegetation-hydrology” to ESM or LSM can significantly improve the simulation accuracy. Therefore, there is a need to critically evaluate the modeling performance of soil-vegetation-hydrological processes in the existing global models and integrate the existing knowledge about the coupling of soil-vegetation-hydrological processes into the global model to build a truly unified understanding of coupled soil-vegetation-hydrological process systems and achieve more accurate simulations at a large scale.

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